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Integrated circuit

Abstract

An integrated circuit includes conductive rails that are disposed in a first conductive layer and separated from each other in a layout view, signal rails disposed in a second conductive layer different from the first conductive layer, at least one first via coupling a first signal rail of the signal rails to at least one of the conductive rails, and at least one first conductive segment. The first signal rail transmits a supply signal through the at least one first via and the at least one of the conductive rails to at least one element of the integrated circuit. The at least one first via and the at least one first conductive segment are disposed above first conductive layer. The at least one first conductive segment is coupled to the at least one of the conductive rails and is separate from the first signal rail.

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Background/Summary

CROSS-REFERENCE (1) The present application is a continuation application of U.S. application Ser. No. 17/868,065, filed Jul. 19, 2022, which is a divisional application of U.S. application Ser. No. 17/035,160, filed Sep. 28, 2020, now U.S. Pat. No. 11,817,392, issued on Nov. 14, 2023, the full disclosures of which are incorporated herein by reference.

BACKGROUND

(1) Integrated circuits (ICs) have experienced exponential growth. The design of the ICs has produced generations with smaller size and having more complicated circuits. Increasingly dense ICs result in benefits in terms of speed, functionality and cost, but cause increasingly difficult design and fabrication issues.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIGS. 1A to 1B are layout diagrams of an integrated circuit (IC), in accordance with some embodiments of the present disclosure.

(3) FIG. 2 is a cross-section schematic diagram of a layout diagram corresponding to the layout diagram of FIG. 1A or 1B, in accordance with some embodiments of the present disclosure.

(4) FIGS. 3A to 3C are layout diagrams of an IC, in accordance with some embodiments of the present disclosure.

(5) FIG. 4 is a layout diagram of an IC, in accordance with some embodiments of the present

disclosure.

(6) FIGS. 5A to 5C are cross-section schematic diagrams of layout diagrams corresponding to the layout diagram of FIG. 4, in accordance with some embodiments of the present disclosure.

(7) FIG. 6 is a flow chart of a method for manufacturing an IC, in accordance with some embodiments of the present disclosure.

(8) FIGS. 7A to 7G are schematic diagrams, in cross-sectional view of part of an IC corresponding to the IC of FIG. 4, illustrating various processes of the method of FIG. 6, in accordance with some embodiments of the present disclosure.

(9) FIG. 8 is a layout diagram of an IC, in accordance with some embodiments of the present disclosure.

(10) FIGS. 9A to 9C are cross-section schematic diagrams of layout diagrams corresponding to the layout diagram of FIG. 8, in accordance with some embodiments of the present disclosure.

(11) FIGS. 10A to 10C are schematic diagrams, in cross-sectional view of part of an IC corresponding to the IC of FIG. 8, illustrating various processes of the method of FIG. 6, in accordance with some embodiments of the present disclosure.

(12) FIG. 11A is a circuit diagram of an IC, in accordance with some embodiments of the present disclosure.

(13) FIG. 11B is a layout diagram of an IC corresponding to the IC of FIG. 11A, in accordance with some embodiments of the present disclosure.

(14) FIG. 12A is a circuit diagram of an IC, in accordance with some embodiments of the present disclosure.

(15) FIG. 12B is a layout diagram of an IC corresponding to the IC of FIG. 12A, in accordance with some embodiments of the present disclosure.

(16) FIG. 13A is a circuit diagram of an IC, in accordance with some embodiments of the present disclosure.

(17) FIG. 13B is a layout diagram of an IC corresponding to the IC of FIG. 13A, in accordance with some embodiments of the present disclosure.

(18) FIG. 14A is a circuit diagram of an IC, in accordance with some embodiments of the present disclosure.

(19) FIG. 14B is a layout diagram of an IC corresponding to the IC of FIG. 14A, in accordance with some embodiments of the present disclosure.

(20) FIG. 15A is a circuit diagram of an IC, in accordance with some embodiments of the present disclosure.

(21) FIG. 15B is a layout diagram of an IC corresponding to the IC of FIG. 15A, in accordance with some embodiments of the present disclosure.

(22) FIG. 16A is a circuit diagram of an IC, in accordance with some embodiments of the present disclosure.

(23) FIG. 16B is a layout diagram of an IC corresponding to the IC of FIG. 16A, in accordance with some embodiments of the present disclosure.

(24) FIG. 17A is a circuit diagram of an IC, in accordance with some embodiments of the present disclosure.

(25) FIGS. 17B to 17E is a layout diagram of an IC corresponding to the IC of FIG. 17A, in accordance with some embodiments of the present disclosure.

(26) FIG. 18A is a circuit diagram of an IC, in accordance with some embodiments of the present disclosure.

(27) FIG. 18B is a layout diagram of an IC corresponding to the IC of FIG. 18A, in accordance with some embodiments of the present disclosure.

(28) FIG. 19A is a circuit diagram of an IC, in accordance with some embodiments of the present disclosure.

(29) FIG. 19B is a layout diagram of an IC corresponding to the IC of FIG. 19A, in accordance

with some embodiments of the present disclosure.

(30) FIG. **20** is a flow chart of a method for fabricating an IC, in accordance with some embodiments of the present disclosure.

(31) FIG. **21** is a block diagram of a system for designing an IC layout design, in accordance with some embodiments of the present disclosure.

(32) FIG. **22** is a block diagram of an IC manufacturing system, and an IC manufacturing flow associated therewith, in accordance with some embodiments.

DETAILED DESCRIPTION

(33) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(34) The terms used in this specification generally have their ordinary meanings in the art and in the specific context where each term is used. The use of examples in this specification, including examples of any terms discussed herein, is illustrative, and in no way limits the scope and meaning of the disclosure or of any exemplified term. Likewise, the present disclosure is not limited to various embodiments given in this specification.

(35) Although the terms “first,” “second,” etc., may be used herein to describe various elements, these elements should not be limited by these terms. These terms are used to distinguish one element from another. For example, a first element could be termed a second element, and, similarly, a second element could be termed a first element, without departing from the scope of the embodiments. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

(36) In this document, the term “coupled” may also be termed as “electrically coupled”, and the term “connected” may be termed as “electrically connected”. “Coupled” and “connected” may also be used to indicate that two or more elements cooperate or interact with each other.

(37) Furthermore, spatially relative terms, such as “underlying,” “below,” “lower,” “overlying,” “upper” and the like, may be used throughout the description for ease of understanding to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The structure may be otherwise oriented (e.g., rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(38) As used herein, “around”, “about”, “approximately” or “substantially” shall generally refer to any approximate value of a given value or range, in which it is varied depending on various arts in which it pertains, and the scope of which should be accorded with the broadest interpretation understood by the person skilled in the art to which it pertains, so as to encompass all such modifications and similar structures. In some embodiments, it shall generally mean within 20 percent, preferably within 10 percent, and more preferably within 5 percent of a given value or range. Numerical quantities given herein are approximate, meaning that the term “around”, “about”, “approximately” or “substantially” can be inferred if not expressly stated, or meaning other approximate values.

(39) Reference is now made to FIGS. **1A** and **1B**. FIG. **1A** is a layout diagram **100A** of an

integrated circuit (IC), and FIG. 1B is a layout diagram **100B** of an IC, in accordance with some embodiments of the present disclosure. For simplicity of illustration of the layout diagram **100A** or **100B**, it is merely illustrates a few patterns disposed in one metal layer which, for example, in some embodiments, is a metal-zero (**M0**) layer hereinafter.

(40) For illustration in FIG. 1A, the layout diagram **100A** includes conductive rails **P01**, **P02**, **S01**, **S02**, **S03** and **S04**, conductive segments **111**, **112**, **121**, **122** and **123**, and cut segments CT. For simplicity, each of the conductive rails **S01**, **S02**, **S03** and **S04** is referenced as **S0** hereinafter for illustration, because each of the conductive rails **S01**, **S02**, **S03** and **S04** operates in a similar way in some embodiments.

(41) The conductive rails **P01**, **P02** and **S0** are disposed in the same metal layer which, in some embodiments, the **M0** layer, and are separated from each other in a layout view. The conductive rails **S0** are disposed between the conductive rails **P10** and **P20** in a layout view. The conductive segment **111** is disposed on the conductive rail **P01**, and the conductive segment **112** is disposed on the conductive rail **P02**. The conductive segments **121-123** are disposed on and between at least two adjacent conductive rails **S0**. Specifically, the conductive segment **121** is disposed on and disposed between the conductive rails **S03** and **S04**. The conductive segment **122** is disposed on and disposed between the conductive rails **S02** and **S03**. The conductive segment **123** is disposed on and disposed between the conductive rails **S01** and **S02**. Alternatively stated, in a layout view, the conductive segment **121** is partially overlapped with the conductive rails **S03** and **S04**; the conductive segment **122** is partially overlapped with the conductive rails **S02** and **S03**; and the conductive segment **123** is partially overlapped with the conductive rails **S01** and **S02**. Each of the cut segments CT is disposed across one of the conductive rails **S0**. For example, with reference to FIG. 1A, the cut segment CT with labeling is disposed across the conductive rail **S02**.

(42) In some embodiments, the conductive rails **P01** and **P02** are configured to receive power signals (which are discussed at least with reference to FIGS. 4 and 5A to 5C) that are disposed in another metal layer which, for example, in some embodiments, is a first metal (**M1**) layer hereinafter. The **M1** layer is disposed above the **M0** layer where disposed the conductive rails **P01** and **P02**. In various embodiments, the conductive rails **P01** and **P02** are indicated as power conductive rails, for transmitting the power signals including, for example, supply voltage signals (which is indicated as VDD discussed with reference to FIG. 4) or reference voltage signals (which is indicated as VSS discussed with reference to FIG. 4). In some other embodiments, the conductive rails **P01** and **P02** are referred to as a pair of power conductive rails, and are configured to transmit power signals with voltages that are different from one another.

(43) In some embodiments, the conductive rails **S0** are configured to receive data signals that are disposed in a same metal layer where the conductive rails **P01** and **P02** are disposed (i.e., the **M0** layer). In various embodiments, the conductive rails **S0** are indicated as signal conductive rails, for transmitting the data signals including, for example, signals (which is indicated as D1 and D2 shown in FIG. 4). In some other embodiments, at least two conductive rails **S0** are coupled to each other, for transmitting the same data signals.

(44) In some embodiments, each of the cut segments CT is configured to cut off at least one pattern in the layout diagram **100A**, for separating this pattern into two portions. In various embodiments, the cut segments CT are removed before generating a layout diagram for fabricating the IC.

(45) In some embodiments, the conductive segments **111** and **112** are shaped as extending rails that are similar to the conductive rails **P01** and **P02**. In various embodiments, the conductive segment **111** is coupled to the conductive rail **P01**, and is spaced apart from the power signal that is disposed in **M1** layer and is coupled to the conductive rail **P01**. In further embodiments, the conductive segment **111** contacts the conductive rail **P01**. Similarly, the conductive segment **112** is coupled to the conductive rail **P02**, and is spaced apart from the power signal that is disposed in **M1** layer and is coupled to the conductive rail **P02**. In further embodiments, the conductive segment **112** contacts the conductive rail **P02**. In various embodiments, the conductive segments **111** and **112** are

indicated as “V0 rail”, for forming additional conductive segments/traces/tracks/patterns on the conductive rails **P01** and **P02**. Alternatively stated, a height of the conductive rails **P01** or **P02** in the **M0** layer is increased by disposing the conductive segments **111** or **112** on and above the conductive rails **P01** or **P02**. In some other embodiments, the conductive rails **P01** and **P02**, and the conductive segments **111** and **112** are made of a metal material that is same as one another.

(46) In some embodiments, the conductive segments **121**, **122** and **123** are shaped as a block. In various embodiments, the conductive segment **121** is coupled between the conductive rails **S03** and **S04**, and is spaced apart from the data signal (which is discussed at least with reference to FIGS. 4 and 5A to 5C) that is disposed in **M1** layer. In further embodiments, the conductive segment **121** contacts the conductive rails **S03** and **S04**. Similarly, the conductive segment **122** is coupled between the conductive rails **S02** and **S03**, and the conductive segment **123** is coupled between the conductive rails **S01** and **S02**. Each of the conductive segments **122** and **123** is spaced apart from the data signal that is disposed in **M1** layer. In further embodiments, the conductive segment **122** contacts the conductive rails **S02** and **S03**; and the conductive segment **123** contacts the conductive rails **S01** and **S02**. In various embodiments, the conductive segments **121**, **122** and **123** are indicated as “**M0** jumper”, for forming additional conductive segments/traces/tracks/patterns on the conductive rails **S0**, and disposing across at least two adjacent conductive rails **S0**. Alternatively stated, a height of the conductive rails **S0** in the **M0** layer is increased by disposing the conductive segments **121**, **122** or **123** on and above the conductive rails **S0**. In some other embodiments, the conductive rails **S0**, and the conductive segments **121**, **122** and **123** are made of a metal material that is same as one another.

(47) The layout diagram **100B** of the IC has a configuration similar to that of the layout diagram **100A** of the IC as illustrated in FIG. 1A, and similar detailed description is therefore omitted. Compared to FIG. 1A, the layout diagram **100B** shown in FIG. 1B includes a conductive segment **124**, rather than the conductive segments **121-123**, disposed on the conductive rails **S0**. The conductive segment **124** is disposed on and disposed between the conductive rails **S02** and **S03**. Alternatively stated, in a layout view, the conductive segment **124** is partially overlapped with the conductive rails **S02** and **S03**. In some embodiments, the conductive segment **124** has a configuration or arrangement similar to the conductive segments **121-123** shown in FIG. 1A.

(48) The number and arrangement of conductive segments or conductive rails shown in FIGS. 1A and 1B are given for illustrative purposes. Various numbers and arrangements of conductive segments or conductive rails to implement the layout diagram **100A** in FIG. 1A or the layout diagram **100B** in FIG. 1B are within the contemplated scope of the present disclosure. For example, in some embodiments, in addition to the conductive segments **121**, **122** and **123** or the conductive rails **P01**, **P02** and **S0** shown in FIG. 1A, the layout diagram **100A** further includes the conductive rails disposed between the conductive rails **P01**, **P02** and next to the conductive rails **S0**, in a layout view.

(49) Reference is now made to FIG. 2. FIG. 2 is a cross-section schematic diagram of a layout diagram **200** corresponding to the layout diagram **100A** of FIG. 1A or the layout diagram **100B** of FIG. 1B, taken along a line **10-10**, in accordance with some embodiments of the present disclosure. The layout diagram **200** with respect to the embodiments of FIG. 1A or 1B, like elements in FIG. 2 are designated with the same reference numbers for ease of understanding.

(50) For illustration in FIG. 2, the conductive segments **211** and **212** are disposed on the conductive rails **P10** and **P20**, respectively, as discussed with embodiments shown in FIGS. 1A and 1B. The conductive segment **213** is disposed on two adjacent conductive rails **S02** and **S03**. Furthermore, a part of the conductive segment **213** is disposed in a space between the conductive rails **S02** and **S03**, in a direction along the line **10-10** which is also referred to as a column direction with reference to FIGS. 1A-1B. Alternatively stated, the conductive segment **213** is disposed stuck between the conductive rails **S02** and **S03**.

(51) Reference is now made to FIGS. 3A to 3C. FIGS. 3A to 3C are layout diagrams **300A**, **300B**

and **300C** of an IC, in accordance with some embodiments of the present disclosure. Each of the layout diagrams **300A**, **300B** and **300C** of the IC has a configuration similar to that of the layout diagram **100A** of the IC as illustrated in FIG. **1A** or the layout diagram **100B** of the IC as illustrated in FIG. **1B**, and similar detailed description is therefore omitted. Each of the layout diagrams **300A**, **300B** and **300C** with respect to the embodiments of FIG. **1A** or **1B**, like elements in FIGS. **3A** to **3C** are designated with the same reference numbers for ease of understanding.

(52) Compared to FIG. **1A** or **1B**, each of the layout diagrams **300A**, **300B** and **300C** shown in FIG. **3A** to **3C** further includes gate segments **311** and **312** disposed across the conductive rails **P01** and **P02** and the conductive rails **S0**, in a layout view. As illustrated in FIGS. **3A** to **3C**, the conductive rails **P01**, **P02** and **S0** extend along a row direction, and the gate segments **311** and **312** extend along a column direction that is substantially perpendicular to the row direction.

(53) In some embodiments, the gate segments **311** and **312** are formed in an active area that is below the **M0** layer, for forming gate terminals of transistors included in the IC. In various embodiments, the gate segments **311** and **312** include polysilicon (PO), and are formed by a gate forming process, which is also referred to as an “PO process” using a polysilicon mask.

(54) In some embodiments, a distance between two adjacent gates which are, for example, with reference to FIGS. **3A** to **3C**, the gate segments **311** and **312**, is indicated as a gate pitch **D1**. The gate pitch **D1** is also referred to as a poly pitch **D1** hereinafter. In some other embodiments, a distance between two adjacent conductive rails which are, for example, with reference to FIGS. **3A** to **3C**, the rail segments **S01** and **S02**, is indicated as a **M0** pitch **D2**. In alternative embodiments, at least one of the poly pitch **D1** or the **M0** pitch **D2** is limited by design rules in the layout diagram of the IC, and the design rules are associated with the fabrication processes or technologies of the IC.

(55) For illustration in FIG. **3A**, the layout diagram **300A** includes conductive segments **121**, **122** and **123** shaped as blocks and disposed on and between the conductive rails **S0**. The layout diagram **300A** also includes a pair of conductive segments **111** and **112** shaped as extending rails and disposed on a pair of conductive rails **P01** and **P02**, respectively. For each of the conductive segments **121**, **122** and **123**, it is shaped as a square, and a width of each of the conductive segments **121**, **122** and **123** is approximately equal to one **M0** pitch **D2**. Alternatively stated, a size of each of the conductive segments **121**, **122** and **123** is approximately equal to a square of **M0** pitch **D2** (i.e., **M0** pitch **D2*****M0** pitch **D2**).

(56) A distance between corresponding corners of two adjacent conductive segments is about in a range of half of one poly pitch **D1** to one poly pitch **D1** (i.e., $0.5 \times \text{poly pitch } D1 \sim 1 \times \text{poly pitch } D1$). As illustrated in FIG. **3A**, a corner of the conductive segments **121** is separated from a corresponding corner of the conductive segments **122** by a distance **C1**, and the distance **C1** is about in a range discussed above.

(57) Furthermore, a width of each of the conductive segments **111** and **112** is substantially equal to or smaller than a width of each of the conductive rails **P01** and **P02**. In some embodiments, a length of each of the conductive segments **111** and **112** is also substantially equal to or smaller than a length of each of the conductive rails **P01** and **P02**.

(58) A distance between one conductive segment indicated as the **V0** rail and one conductive segment indicated as the **M0** jumper is about in a range of half of the poly pitch **D1** to 1.2 times of the poly pitch **D1** (i.e., $0.5 \times \text{poly pitch } D1 \sim 1.2 \times \text{poly pitch } D1$), in some embodiments. As illustrated in FIG. **3A**, the conductive segment **111** is indicated as the **V0** rail that is disposed on one of the conductive rails **P01** and **P02**. The conductive segment **123** is indicated as the **M0** jumper that is disposed on at least two adjacent conductive rails **S0**. The conductive segment **111** is separated from the conductive segment **123**, which is disposed adjacent to the conductive segment **111**, by a distance **C2**. The distance **C2** is about in a range discussed above.

(59) Compared to embodiments illustrated in FIG. **3A**, the layout diagram **300B** includes one conductive segment **124** shaped as a railed block and disposed on and between the conductive rails **S0**. The conductive segment **124**, it is shaped as a rectangular block, and a length **L1** of the

conductive segment **124** is greater than twice of the poly pitch **D1** (i.e., $L1 > 2 * \text{poly pitch D1}$), in some embodiments.

(60) A distance between one conductive segment indicated as the **V0** rail and one conductive segment indicated as the **M0** jumper is about in a range of half of the poly pitch **D1** to one of the poly pitch **D1** (i.e., $0.5 * \text{poly pitch D1} \sim 1 * \text{poly pitch D1}$), in some embodiments. For example, with reference to FIG. 3B, the conductive segment **111** disposed the conductive rail **P01** is separated from the conductive segment **124**, that is disposed adjacent to the conductive segment **111** and disposed on the conductive rail **S01**, by a distance **C3**. The distance **C3** is about in a range discussed above.

(61) Compared to embodiments illustrated in FIG. 3A, the layout diagram **300C** further includes one conductive segment **124** shaped as a railed block and disposed on and between the conductive rails **S0**. In some embodiments, the layout diagram **300C** is an alternative embodiment combining with the layout diagrams **300A** and **300B**.

(62) Each of the conductive segments **121**, **122** and **123** is shaped as a square, and a width of the same is approximately equal to one **M0** pitch **D2**. Alternatively stated, a size of each of the conductive segments **121**, **122** and **123** is approximately equal to a square of **M0** pitch **D2** (i.e., $\text{M0 pitch D2} * \text{M0 pitch D2}$). Similar to that illustrated in FIG. 3A, a distance between two adjacent conductive segments **121**, **122** or **123** is about in a range of half of one poly pitch **D1** to one poly pitch **D1** (i.e., $0.5 * \text{poly pitch D1} \sim 1 * \text{poly pitch D1}$). For example, with reference to FIG. 3C, a distance between a corner of the conductive segments **121** and a corresponding corner of the conductive segment **122** is indicated as a distance **C4**. The distance **C4** is in a range of $0.5 * \text{poly pitch D1}$ to $1 * \text{poly pitch D1}$. Furthermore, a length or width of the conductive segments **111** or **112** is also substantially equal to or smaller than a length or a width of the conductive rails **P01** or **P02**.

(63) In some embodiments, a distance between one conductive segment indicated as the **V0** rail and another conductive segment indicated as the **M0** jumper is not limited in the layout diagram **300C**. For example, with reference to FIG. 3C, a distance between the conductive segment **111** indicated as the **V0** rail and the conductive segment **124** indicated as the **M0** jumper may be smaller than $0.5 * \text{poly pitch D1}$.

(64) In some embodiments, the layout diagram **300C** is generated by at least one mask (which is discussed with reference to FIG. 22). For example, the conductive segments **111** and **112** are patterned by utilizing a first mask. The remaining conductive segments **121**, **122** and **123** are patterned by utilizing a second mask. Accordingly, the layout diagram **300C** has less design limitations compared to the layout diagram **300A** or **300B** that is patterned by one mask.

(65) The configuration and arrangement of the layout diagrams **300A**, **300B** and **300C** shown in FIGS. 3A to 3C is given for illustrative purposes. Various configurations and arrangements of the layout diagrams **300A**, **300B** and **300C** to implement the IC are within the contemplated scope of the present disclosure. For example, in some embodiments, in a layout view, at least one of the conductive segments **121**, **122**, **123** or **124** is disposed on and between or across more than three conductive rails **S0**.

(66) Reference is now made to FIG. 4. FIG. 4 is a layout diagram **400** of an IC, in accordance with some embodiments of the present disclosure. The layout diagram **400** with respect to the embodiments of FIG. 1A or 1B, like elements in FIG. 4 are designated with the same reference numbers for ease of understanding.

(67) For illustration in FIG. 4, the layout diagram **400** includes conductive rails **P01**, **P02** and **P03**, and conductive rails **S01**, **S02**, **S03** and **S04** disposed in the **M0** layer and extending along a row direction. In some embodiments, the conductive rails **S01**, **S02**, **S03** and **S04** are referenced as **S0** hereinafter for simplicity of illustration. In a layout view, the conductive rails **P01** and **P02** are separated from one another by the conductive rails **S0**, and the conductive rails **P02** and **P03** are separated from one another by additional conductive rails that are not shown in FIG. 4. In some embodiments, the conductive rail **P01** is identical to the conductive rail **P03**. The conductive rails

P01, **P02**, and **S0** of the layout diagram **400** have configurations similar to that of the layout diagram as illustrated in FIG. 1A, and similar detailed description is therefore omitted.

(68) The layout diagram **400** further includes conductive segments **411a**, **411b**, **412a**, **412b** and **421** disposed on the conductive rails **P01**, **P02**, and **S0**, signal rails **P11**, **P12**, **S11** and **S12** disposed in the **M1** layer, and vias **431**, **432**, **433** and **434** disposed between the **M0** layer and the **M1** layer. The signal rails **P11**, **P12**, **S11** and **S12** are separated from each other, and extend along a column direction, in a layout view. Alternatively stated, in a layout view, the signal rails **P11**, **P12**, **S11** and **S12** are disposed above and across the conductive rails **P01**, **P02**, **P03** and **S0**. The vias **431**, **432**, **433** and **434** are overlapped with at least two of the conductive rails **P01**, **P02** and **S0** and the signal rails **P11**, **P12**, **S11** and **S12**, in a layout view. Specifically, in a layout view, the via **431** is overlapped with the conductive rail **P01** and the signal rail **P11**; the via **434** is overlapped with the conductive rail **P02** and the signal rail **P12**; the via **432** is overlapped with the conductive rail **S01** and the signal rail **S11**; and the via **433** is overlapped with the conductive rail **S03** and the signal rail **S12**.

(69) The conductive segments **411a** and **411b** are disposed on the conductive rail **P01** and are separated from each other, in a layout view. The via **431** is disposed between the conductive segments **411a** and **411b**, in a layout view. Alternatively stated, one conductive segment indicated as the **V0** rail is disposed on the conductive rail **P01**. Such conductive segment includes several separated portions including, for example, with reference to FIG. 4, the conductive segments **411a** and **411b**. In addition, at least one via is disposed between these separated portions, in a layout view. With reference to FIG. 4, the via **431** is disposed between the separated portions which are the conductive segments **411a** and **411b**, in a layout view. Similarly, the conductive segments **412a** and **412b** are disposed on the conductive rail **P02**, and are separated from each other in a layout view. The via **434** is disposed between the conductive segments **412a** and **412b**, in a layout view.

(70) The conductive segment **421** is disposed on and disposed between the conductive rails **S02** and **S03**. The conductive segment **421** is shaped as a railed block and extends along the row direction. Alternatively stated, at least one conductive segment indicated as the **M0** jumper extends parallel to the conductive rails **S0**. Such conductive segment is disposed on and between the conductive rails **S0**, and is separated from another conductive segment indicated as the **V0** rail in a layout view, in some embodiments. For example, the conductive segment **421** that is indicated as the **M0** jumper is separated from the conductive segment **411b** or **412b** that is indicated as the **V0** rail, in a layout view.

(71) In some embodiments, the signal rails **P11** and **P12** are configured to provide power signals to the conductive rails **P01** and **P02**. In various embodiments, the signal rail **P11** is configured to provide a voltage signal that is different from that of the signal rail **P12**. For example, the signal rail **P11** is configured to provide the supply voltage signal with voltage **VDD**, and the signal rail **P12** is configured to provide the reference voltage signal with voltage **VSS**. The voltage **VDD** is higher than the voltage **VSS** which, in some embodiments, is referred to as a ground. In alternative embodiments, the signal rails **P11** and **P12** are indicated as power rails, for providing the power signals (e.g., **VDD** or **VSS**) to the power conductive rails including, for example, the conductive rails **P01**, **P02** and **P03**.

(72) In some embodiments, the signal rails **S11** and **S12** are configured to provide data signals to the conductive rails **S0**. In various embodiments, the signal rail **S11** is configured to provide a data signal that is different from that of the signal rail **S12**. For example, the signal rail **S11** is configured to provide a control signal for controlling first transistors, and the signal rail **S12** is configured to provide another control signal for controlling second transistors. In alternative embodiments, the signal rails **S11** and **S12** are indicated as data rails, for providing the data signals to the signal conductive rails including, for example, the conductive rails **S01** to **S04**.

(73) Reference is now made to FIGS. 5A to 5C. FIG. 5A is a cross-section schematic diagram of a layout diagram **500A** corresponding to the layout diagram **400** of FIG. 4, taken along a line A-A';

FIG. 5B is a cross-section schematic diagram of a layout diagram **500B** corresponding to the layout diagram **400** of FIG. 4, taken along a line B-B'; and FIG. 5C is a cross-section schematic diagram of a layout diagram **500C** corresponding to the layout diagram **400** of FIG. 4, taken along a line C-C', in accordance with some embodiments of the present disclosure. The layout diagrams **500A** to **500C** with respect to the embodiments of FIG. 4, like elements in FIGS. 5A to 5C are designated with the same reference numbers for ease of understanding.

(74) For illustration in FIG. 5A, the signal rail **P11** extends along the line A-A' and across from the conductive rails **P01**, **S0** to the conductive rail **P02**. The via **431** is disposed between the conductive rail **P01** and the signal rail **P11** without extra spacing. The conductive segment **412a** is disposed on the conductive rail **P02** and is spaced apart from the signal rail **P11**. With above configurations, a height of the conductive segment **412a** is smaller than a height of the via **431**.

(75) In some embodiments, the conductive rail **P01** is coupled through the via **431** to the signal rail **P11**, for receiving the supply voltage signal with voltage VDD. With the above configurations, since the conductive segment **412a** is separated from the signal rail **P11**, the conductive segment **412a** or the conductive rail **P02** is not coupled to or does not contact the signal rail **P11**. In various embodiments, both of the conductive segment **412a** and the conductive rail **P02** are coupled to the ground by separating from the signal rail **P11**. In addition, the conductive rail **P02** is not coupled to any signal rails disposed in the **M1** layer, and is indicated as a ground rail. Similarly, since the conductive rails **S0** are separated from the signal rail **P11**, these conductive rails **S0** are not coupled to the signal rail **P11**.

(76) For illustration in FIG. 5B, the signal rail **S11** extends along the line B-B' and across from the conductive rails **P01**, **S0** to the conductive rail **P02**. The conductive segment **411b** is disposed on the conductive rail **P01** and is spaced apart from the signal rail **S11**. Similarly, the conductive segment **412b** is disposed on the conductive rail **P02** and is spaced apart from the signal rail **S11**. The conductive segment **421** is disposed on two adjacent conductive rails **S02** and **S03** and is also spaced apart from the signal rail **S11**. Alternatively stated, each of the conductive segments **411b**, **412b** and **421** is separated from the signal rail **S11** along a vertical direction that is perpendicular to the line B-B' illustrated in FIG. 5B. In addition, the via **432** is disposed between the conductive rail **S01** and the signal rail **S11** without extra spacing. With above configurations, a height of the conductive segment **411b**, **421** or **412b** is smaller than a height of the via **432**.

(77) Compared to embodiments illustrated in FIG. 5A, the layout diagram **500B** further illustrates conductive segments **511** and **512**, and vias **521** and **522**. The conductive segments **511** and **512** are disposed in the active area (not shown) that is below the **M0** layer, and the vias **521** and **522** are disposed between the active area and the **M0** layer. Specifically, the via **521** is disposed between the conductive segment **511** and the conductive rail **S01** without extra spacing. The via **522** is disposed between the conductive segment **512** and the conductive rail **S03** without extra spacing.

(78) In some embodiments, the conductive segment **511** is coupled through the via **521** to the conductive rail **S01**. The conductive rail **S01** is further coupled through the via **432** to the signal rail **S11**, for receiving the data signal provided to the conductive segment **511**. Furthermore, the conductive segment **512** is coupled through the via **522** to the conductive rail **S03**, and the conductive rail **S03** is further coupled through the conductive segment **421** to the conductive rail **S02**. Alternatively stated, at least two adjacent conductive rails including, for example, the conductive rails **S02** and **S03**, are coupled together through the conductive segment indicated as the **M0** jumper including, for example, the conductive segment **421**.

(79) In some embodiments, the conductive segments **511** and **512** are disposed in the active area, for forming source or drain terminals of transistors included in the IC. In various embodiments, the conductive segments **511** and **512** are referred to as a metal-like defined (MD) segments hereinafter. In alternative embodiments, the vias **521** and **522**, coupled from the MD segments **511** and **512** to other elements of the IC, are referred to as via-defined (VD) vias hereinafter.

(80) In some embodiments, the MD segment includes a portion of at least one metal layer, e.g., one

or more of copper (Cu), silver (Ag), tungsten (W), titanium (Ti), nickel (Ni), tin (Sn), aluminum (Al) or another metal or material suitable for providing a low resistance electrical connection between elements included in the IC, i.e., a resistance level below a predetermined threshold corresponding to one or more tolerance levels of a resistance-based effect on circuit performance. In some other embodiments, the MD segment includes a section of the semiconductor substrate and/or an epitaxial layer having a doping level, e.g., based on an implantation process, sufficient to cause the segment to have the low resistance level. In various embodiments, a doped MD segment includes one or more of silicon (Si), SiGe, silicon-carbide (SiC), boron (B), phosphorous (P), arsenic (As), gallium (Ga), a metal as discussed above, or another material suitable for providing the low resistance level. In some embodiments, an MD segment includes a dopant having a doping concentration of about 1×10^{16} per cubic centimeter (cm.sup.-3) or greater.

(81) For illustration in FIG. 5C, the signal rail **S12** extends along the line C-C' and across from the conductive rails **P01**, **S0** to the conductive rail **P02**. The conductive segment **411b** is disposed on the conductive rail **P01** and is spaced apart from the signal rail **S12**. Similarly, the conductive segment **412b** is disposed on the conductive rail **P02** and is spaced apart from the signal rail **S12**. Alternatively stated, each of the conductive segments **411b** and **412b** is separated from the signal rail **S12** along a vertical direction that is perpendicular to the line C-C' illustrated in FIG. 5C. In addition, the via **433** is disposed between the conductive rail **S03** and the signal rail **S12** without extra spacing. With above configurations, a height of the conductive segment **411b** or **412b** is smaller than a height of the via **433**.

(82) In some embodiments, the conductive rail **S03** is coupled through the via **433** to the signal rail **S12**, for receiving the data signal provided to the conductive rail **S03**. The conductive rail **P01** or **P02** is not coupled to the signal rail **P12**, for receiving the signal that is equivalently indicated as the ground.

(83) Compared to embodiments illustrated in FIG. 5A, the layout diagram **500C** further illustrates MD segments **513** and **514**, and VD vias **523** and **524**. The via **523** is disposed between the MD segment **513** and the conductive rail **S01** without extra spacing, and the via **524** is disposed between the MD segment **514** and the conductive rail **P02** without extra spacing. The MD segments **513** and **514** or the vias **523** and **524** have configurations similar to that of the MD segments **511** and **512** or the vias **521** and **522**, correspondingly, as illustrated in FIG. 5B, and similar detailed description is therefore omitted.

(84) In some approaches, no other conductive segments disposed on the conductive rails in the **M0** layer. With such configuration, a resistance coupled between the conductive rails and the signal rails in the **M1** layer is substantially contributed to a height of the conductive rail and a size of the via coupled between the **M0** and **M1** layers. It affects IR-drop issue and further slows down a working speed of the IC.

(85) Compared to the above approaches, additional conductive segment including, for example, the conductive segment **412a**, is disposed on the conductive rails including, for example, the conductive rail **P02**, in the **M0** layer. This additional conductive segment **412a** is spaced apart from the **M1** layer, for increasing a height of the conductive rail **P02** in the **M0** layer without changing a cell height of the corresponding layout diagram **400**. Accordingly, a resistance of the conductive rail **P02** that is also referred to as the power conductive rail is reduced. It is reduced by disposing the conductive segment **412a** indicated as the **V0** rail on the top surface of the conductive rail **P02**.

(86) Furthermore, additional conductive segment including, for example, the conductive segment **421**, is disposed on and between at least two adjacent conductive rails including, for example, the conductive rails **S02** and **S03**. This additional conductive segment **421** is also spaced apart from the **M1** layer, for coupling these two adjacent conductive rails **S02** and **S03** together as a local interconnection. Accordingly, routings for coupling more than two conductive rails **S0** are reduced by disposing the conductive segment **421** that is indicated as the **M0** jumper. It further improves the electromagnetic bottleneck of the data signals transmitted in these conductive rails **S0** and the IC

performance.

(87) Reference is now made to FIG. 6 and FIGS. 7A to 7G. FIG. 6 is a flow chart of a method **600** for manufacturing an IC including the conductive rails **P01**, **P02** and **S0**, the conductive segments **411a**, **411b**, **412a**, **412b** and **421**, and vias **431** to **434** shown in FIG. 4, or an IC shown in FIGS. 7A to 7G, in accordance with some embodiments of the present disclosure. FIGS. 7A to 7G are schematic diagrams, in cross-sectional view of part of the IC **700** corresponding to the IC of FIG. 4 along the column direction as illustrated in FIG. 4, illustrating various processes of the method **600** of FIG. 6, in accordance with some embodiments of the present disclosure. It is understood that additional operations can be provided before, during, and after the processes shown by FIG. 6 and FIGS. 7A to 7G, and some of the operations described below can be replaced or eliminated, for additional embodiments of the method. The order of the operations/processes may be interchangeable. Throughout the various views and illustrative embodiments, like reference numbers are used to designate like elements.

(88) In operation **S610** of FIG. 6, multiple conductive rails are formed within the **M0** layer of the IC **700**. The conductive rails include the conductive rails **P01**, **P02** and **S0** illustrated in FIG. 4, and dielectric material stack between these conductive rails.

(89) In operation **S620**, a film structure is formed on the conductive rails. Accordingly, the film structure is formed on the **M0** layer of the IC **700**. Alternatively stated, the film is deposited on the **M0** layer. The operations **S610**-**S620** are performed as illustrated in FIG. 7A.

(90) For illustration, as shown in FIG. 7A, the IC **700** includes conductive rails **P02**, **S02**, **S03** and **S04**, and dielectric structure **710** formed in the **M0** layer, and includes a film structure **720** formed on the **M0** layer. The conductive rails **P02**, **S02**, **S03** and **S04** are separated from each other by the dielectric structure **710**. In some embodiments, a material of the dielectric structure **710** is different from a material of the film structure **720**. In various embodiments, the dielectric structure **710** and the film structure **720** are isolated from each other. In alternative embodiments, a thickness of the film structure **720** is substantially equal to a distance between the **M0** layer and the **M1** layer (which is at least shown in FIG. 7E).

(91) In operation **S630**, the film structure is patterned to form patterns. In some embodiments, the patterning operation is utilized photolithography and etching, for spacing a specific pattern in the film structure. The operation **S630** is processed as illustrated in FIG. 7B.

(92) For illustration, as shown in FIG. 7B, patterns **731** and **732** are formed in the film structure **720**. Specifically, the pattern **731** is formed in the film structure **720** and formed above and between the conductive rails **S03** and **S04**. At a surface of the **M0** layer and a bottom of the film structure **720**, a part of the conductive rails **S03** and **S04** and a part of the film structure **720** are removed to form the pattern **731**. Moreover, the pattern **732** is formed in the film structure **720** and formed above and between the conductive rail **P02**. Similarly, around the surface of the **M0** layer, part of the film structure **720** is removed to form the pattern **732**.

(93) In operation **S640**, the patterns are filled with a conductive material, to form a first conductive structure that includes a first conductive segment contacting a first conductive rail of the conductive rails. In some embodiments, the first conductive segment corresponds to at least one of the conductive segment **411a** or **411b** illustrated in FIG. 4 and the first conductive rail corresponds to the conductive rail **P01** illustrated in FIG. 4. In other embodiments, the first conductive segment corresponds to at least one of the conductive segment **412a** or **412b** illustrated in FIG. 4 and the first conductive rail corresponds to the conductive rail **P02** illustrated in FIG. 4. The operation **S640** is processed as illustrated in FIGS. 7C-7D.

(94) In some embodiments, the filling operation in **S640** is also indicated as a multi-metal gap fill process. In various embodiments, the patterns are filled with the conductive material, for forming interconnect structures between two adjacent metal layers which, for example, are the **M0** layer and the **M1** layer.

(95) For illustration, as shown in FIG. 7C, a conductive structure **740** is formed on the film

structure **720**, and is further filled in the patterns **731** and **732**.

(96) In some embodiments, a material of the conductive structure **740** is the same as a material of the conductive segments **S02** to **S04** and **P02** formed in the **M0** layers. Alternatively stated, a material of the pattern **731** or **732** is the same as a material of the conductive segment **S04** or **P02**. In various embodiments, if the conductive structure **740** physically contacts the conductive segments **S02** to **S04** and **P02** formed in the **M0** layers, the conductive structure **740** and the conductive segments **S02** to **S04** and **P02** are coupled to each other. In alternative embodiments, the conductive material is copper (Cu), cobalt (Co), tungsten (W), Ruthenium (Ru), aluminum (Al), graphene, or any other suitable conductive material.

(97) For illustration, as shown in FIG. 7D, conductive segments **751** and **752** are formed. In some embodiments, the conductive segments **752** corresponds to at least one of the conductive segment **411a**, **411b**, **412a** or **412b** that contacts the conductive rail **P01** or **P02**, illustrated in FIG. 4.

Thereby, the conductive segment **752** is able to form the first conductive segment that contacts the first conductive rail, as discussed in the operation **S640**. In other embodiments, the conductive segments **751** corresponds to the conductive segment **421** that contacts two adjacent conductive rails **S0**, illustrated in FIG. 4.

(98) As illustrated in FIG. 7D, the conductive segment **751** is formed on the **M0** layer, and is formed on the part of the conductive rail **S03**, the film structure **720**, and the part of the conductive rail **S04**. Alternatively stated, the conductive segment **751** is formed above and between the conductive rails **S03** and **S04**. In addition, conductive segment **752** is formed on the **M0** layer, and is also formed on the conductive rail **P02**. Furthermore, the conductive segments **751** and **752**, and the film structure **720** have a same surface that is substantially parallel to the surface of the **M0** layer. Alternatively stated, all of the conductive segments **751** and **752** and the film structure **720** form a substantially flat surface above the **M0** layer. In some embodiments, the conductive segments **751** and **752** are made of the same conductive material as the conductive structure **740** is made.

(99) In some embodiments, the operation **S640** further includes the following operations. The patterns filled with the conductive material and the film structure are polished. In some other embodiments, the polishing operation in **S640** is performed by chemical mechanical polishing (CMP) to remove extra conductive material on the top of surface. In various embodiments, part of the structures is removed with a combination of CMP and dry etch.

(100) In operation **S650**, a dielectric structure covering the first conductive segment is formed. In some embodiments, the dielectric structure is formed and stacked on the remaining structure after performing the operation **S640**. In other embodiments, the **M1** layer is generated by forming the dielectric structure on the film structure, and is processed as illustrated in FIG. 7E.

(101) For illustration, as shown in FIG. 7E, a dielectric structure **760** is generated by forming on the film structure **720** and the conductive segments **751** and **752**. In some embodiments, a material of dielectric structure **710** is the same as a material of the dielectric structure **760**. In various embodiments, a thickness of the dielectric structure **760** is substantially equal to a distance between the **M1** layer and a second metal layer (**M2**, not shown) above the **M1** layer.

(102) In operation **S660**, part of the film structure and part of the dielectric structure are removed, to expose part of a second conductive rail of the conductive rails. In some embodiments, the second conductive rail corresponds to one of the conductive rails **S0** illustrated in FIG. 4.

(103) In operation **S670**, the conductive material is filled in the removed part of the film structure and the removed part of the dielectric structure, to form a second conductive structure that includes a via contacting the exposed part of the second conductive rail and a signal rail contacting the via. In some embodiments, the via included in the second conductive structure corresponds to the via **432** illustrated in FIG. 4, and the signal rail included in the second conductive structure corresponds to the data rail **S11** illustrated in FIG. 4. The operations **S660-S670** are processed as illustrated in FIGS. 7F-7G.

(104) For illustration, as shown in FIG. 7F, part of the film structure **720** and part of the dielectric structure **760** are removed, and a structure **781** is exposed. Furthermore, the conductive material (not shown) are filled in the removed structures **720** and **760**, to form a via **771** and the structure **781** which corresponds to a signal rail. The via **771** and the structure **781** are contacted to each other, and correspond to the via **432** and the signal rail **S11** respectively, as shown in FIG. 4, in some embodiments. Also illustrated in FIG. 7G, part of the film structure **720** and part of the dielectric structure **760** are removed, and a structure **782** is exposed. Furthermore, the removed structures **720** and **760** are filled in the conductive material (not shown), to form a via **772** and the structure **782** which corresponds to a signal rail. The via **772** and the structure **782** are contacted to each other.

(105) As illustrated in FIG. 7F, the IC **700** further includes via **771** formed on the **M0** layer and connected to a structure **781** formed in the **M1** layer. One terminal of the via **771** is formed on the conductive rail **S02**, and the other terminal of the via **771** is formed in the **M1** layer. With the operations discussed above, a height of via **771** is substantially equal to or slightly greater than a distance between the conductive rail **S02** and the structure **781**. Alternatively stated, a height of via **771** is substantially equal to or slightly greater than a distance between the **M0** layer and the **M1** layer.

(106) In some embodiments, the via **771** is configured to be coupled between the conductive rail **S02** and the structure **781** formed in the **M1** layer. In various embodiments, the structure **781** is a signal rail that is configured to providing data signals. In some other embodiments, the via **771** is coupled between the conductive rail **S02** and the data rail **S11**, and corresponds to the via **432** shown in FIG. 4 or 5B.

(107) Moreover, the conductive segment **751** is formed on the **M0** layer as discussed above, and is separated from the **M1** layer by the dielectric structure **760**. Alternatively stated, the conductive segment **751** is spaced apart from the **M1** layer. With the operations discussed above, a height of the conductive segment **751** is smaller than a height of the via **771**. Similar to configurations of the conductive segment **751**, the conductive segment **752** is formed on the **M0** layer and is spaced apart from the **M1** layer. Also, a height of the conductive segment **752** is smaller than a height of the via **771**.

(108) In some embodiments, the conductive segment **751** is coupled between the conductive rails **S03** and **S04**. In some other embodiments, the conductive segment **751** is referred to as the **M0** jumper, and corresponds to the conductive segment **421** shown in FIG. 4 or 5B. In various embodiments, the conductive segment **752** is coupled to the conductive rail **P02**. In some other embodiments, the conductive segment **752** is referred to as the **V0** rail, and corresponds to the conductive segment **412a** shown in FIG. 4 or 5A.

(109) In some embodiments, the operation for forming the conductive segments **751** and **752** and the via **771** with the above arrangement between the **M0** and **M1** layers is referred to as a dual damascene process. It makes connections between the conductive segments **751** and **752** and the structure **781** formed in the **M1** layer. In some other embodiments, the operation for forming the conductive segments **751** and **752** and the via **771** with the above arrangement between the **M0** and **M1** layers is referred to as a single damascene process. Compared to embodiments performed by the dual damascene process, the via **771** performed by the single damascene process has a greater height.

(110) For illustration, as shown in FIG. 7G, the IC **700** includes via **772** formed on the **M0** layer and connected to a structure **782** formed in the **M1** layer. One terminal of the via **772** is formed on the conductive rail **P02**, and the other terminal of the via **772** is formed in the **M1** layer. With the operations discussed above, similar to configurations of the via **771**, a height of via **772** is substantially equal to or slightly greater than a distance between the conductive rail **P02** and the structure **782**.

(111) In some embodiments, the via **772** is configured to couple between the conductive rail **P02**

and the structure **782** formed in the **M1** layer. In various embodiments, the structure **782** is a power rail that is configured to providing power signals. In some other embodiments, the via **772** is coupled between the conductive rail **P02** and the power rail **P11**, and corresponds to the via **431** shown in FIG. **4** or **5A**.

(112) As described above, the integrated circuit in the present disclosure is provided with conductive segments formed on the **M0** layer and is spaced apart from the **M1** layer. The conductive segment indicated as the **V0** rail is disposed on the power conductive rail, and the conductive segment indicated as the **M0** jumper is disposed on at least two signal conductive rails. Accordingly, routing between the **M0** layer and the **M1** layer is reduced, and resistances of the power conductive rails or the signal conductive rails are also reduced by additional arrangements of the conductive segment.

(113) The configuration of FIGS. **7A-7G** is given for illustrative purposes. Various configurations of the elements mentioned above in FIGS. **7A-7G** are within the contemplated scope of the present disclosure.

(114) Reference is now made to FIG. **8**. FIG. **8** is a layout diagram **800** of an IC, in accordance with some embodiments of the present disclosure. The layout diagram **800** has configurations similar to that of the layout diagram **400** as illustrated in FIG. **4**, and similar detailed description is therefore omitted. The patterns shown in the layout diagram **800** with respect to the embodiments of FIG. **4**, like elements in FIG. **8** are designated with the same reference numbers for ease of understanding.

(115) Compared to embodiments illustrated in FIG. **4**, the layout diagram **800** includes conductive segments **811** and **812** disposed on the conductive rails **P01** and **P02** respectively. The layout diagram **800** also includes conductive segment **821** disposed on and between the conductive rails **S02** and **S03**. The layout diagram **800** further includes vias **831**, **832** and **433** disposed between the **M0** layer and the **M1** layer. Specifically, in a layout view, the via **831** is overlapped with the conductive rail **P01**, the conductive segment **811** and the signal rail **P11**; the via **832** is overlapped with a conductive rail **S02**, the conductive rail **S03**, the conductive segment **821** and the signal rail **S11**; and the via **433** has similar configuration to that is shown in FIG. **4**.

(116) In some embodiments, the conductive segment **811** corresponds to the conductive segment **411a** or **411b** shown in FIG. **4**. Compared to the conductive segment **411a** or **411b** of FIG. **4**, the conductive segment **811** is not separated into multiple portions. Similarly, the conductive segment **812** corresponds to the conductive segment **412a** or **412b** shown in FIG. **4**. In various embodiments, the conductive segment **811** has similar configuration/arrangement to the conductive segment **812**. In some other embodiments, the conductive segment **821** corresponds to the conductive segment **421** shown in FIG. **4**, and has similar configuration/arrangement to the conductive segment **421**.

(117) Reference is now made to FIGS. **9A to 9C**. FIG. **9A** is a cross-section schematic diagram of a layout diagram **900A** corresponding to the layout diagram **800** of FIG. **8**, taken along a line A-A'; FIG. **9B** is a cross-section schematic diagram of a layout diagram **900B** corresponding to the layout diagram **800** of FIG. **8**, taken along a line B-B'; and FIG. **9C** is a cross-section schematic diagram of a layout diagram **900C** corresponding to the layout diagram **800** of FIG. **8**, taken along a line C-C', in accordance with some embodiments of the present disclosure. The layout diagram **900A** has configurations similar to that of the layout diagram **500A** as illustrated in FIG. **5A**; the layout diagram **900B** has configurations similar to that of the layout diagram **500B** as illustrated in FIG. **5B**; and the layout diagram **900C** has similar configuration/arrangement to the layout diagram **500C** shown in FIG. **5C**, and similar detailed description is therefore omitted. The layout diagrams **900A to 900C** with respect to the embodiments of FIG. **8**, like elements in FIGS. **9A to 9C** are designated with the same reference numbers for ease of understanding.

(118) For illustration in FIG. **9A**, the conductive segment **811** is disposed on the conductive rail **P01**. The via **831** is disposed between the conductive segment **811** and the signal rail **P11** without

extra spacing. The conductive segment **812** is disposed on the conductive rail **P02** and is spaced apart from the signal rail **P11**.

(119) In some embodiments, the conductive rail **P01** is coupled to the conductive segment **811**, and is further coupled through the via **831** to the signal rail **P11**, for receiving the supply voltage signal with voltage **VDD**. With the above configurations, since the conductive segment **812** is separated from the signal rail **P11**, the conductive segment **812** or the conductive rail **P02** is not coupled to the signal rail **P11**.

(120) For illustration in FIG. **9B**, the conductive segment **811** is disposed on the conductive rail **P01** and is spaced apart from the signal rail **S11**. The conductive segment **812** is disposed on the conductive rail **P02** and is spaced apart from the signal rail **S11**. The conductive segment **821** is disposed on two adjacent conductive rails **S02** and **S03**. In addition, the via **832** is disposed between the conductive segment **821** and the signal rail **S11** without extra spacing.

(121) Compared to embodiments illustrated in FIG. **9A**, the layout diagram **900B** further illustrates conductive segments **911** and **912**, and vias **921** and **922**. In some embodiments, the conductive segments **911** and **912** correspond to the MD segments **511** and **512** in FIG. **5B** respectively, and the vias **921** and **922** correspond to the VD vias **521** and **522** in FIG. **5B** respectively. The conductive segments **911** and **912** are indicated as the MD segments hereinafter, and the vias **921** and **922** are indicated as the VD vias hereinafter.

(122) With reference to FIG. **9B**, the VD via **921** is disposed between the MD segment **911** and the conductive rail **S01** without extra spacing. The VD via **922** is disposed between the MD segments **912** and the conductive rail **S03** without extra spacing.

(123) In some embodiments, the MD segment **911** is coupled through the VD via **921** to the conductive rail **S01**. Furthermore, the MD segment **912** is coupled through the VD via **922** to the conductive rail **S03**. The conductive rail **S03** is further coupled to the conductive segment **821**, and is further coupled through the via **832** to the signal rail **S11**, for receiving the data signal provided to the conductive rail **S03** and further to the MD segment **912**. On the other hand, the conductive rail **S03** is also coupled through the conductive segment **821** to the conductive rail **S02**. Accordingly, the conductive rail **S02** is also configured to receive the data signal transmitted from the signal rail **S11**.

(124) For illustration in FIG. **9C**, conductive segments **913** and **914**, and vias **923** and **924** are disposed below the conductive rails **P01**, **P02** and **S0**. In some embodiments, the conductive segments **913** and **914** correspond to the MD segments **513** and **514** in FIG. **5C** respectively, and the vias **923** and **924** correspond to the VD vias **523** and **524** in FIG. **5C** respectively.

(125) Reference is now made to FIGS. **10A** to **10C** and back to FIG. **6**. FIGS. **10A** to **10C** are schematic diagrams, in cross-sectional view of part of the IC **1000** corresponding to the IC of FIG. **8** along the column direction as illustrated in FIG. **8**, illustrating some processes of the method of FIG. **6**, in accordance with some embodiments of the present disclosure. FIG. **10A** illustrates the operation **S650** in FIG. **6**, and has configurations similar to that is illustrated in FIG. **7E**. FIG. **10B** illustrates the operations **S660-S670** in FIG. **6**, and have configurations similar to that are illustrated in FIGS. **7F** and **7G**. As such, similar detailed description is therefore omitted.

(126) As illustrated in FIG. **10A**, a conductive segment **1051** is disposed on the part of the conductive rail **S03**, the film structure **720** and the part of the conductive rail **S04**, and is disposed on and above the **M0** layer. A conductive segment **1052** is disposed on the conductive rail **P02** and is disposed on and above the **M0** layer. Similar to the conductive segments **751** and **752** shown in FIG. **7E**, the conductive segments **1051** and **1052** and the film structure **720** form a substantially flat surface above the **M0** layer. In some embodiments, the conductive segments **1052** corresponds to at least one of the conductive segment **811** or **812** that contacts the conductive rail **P01** or **P02**, illustrated in FIG. **8**. In other embodiments, the conductive segments **1051** corresponds to the conductive segment **821** that contacts two adjacent conductive rails **S0**, illustrated in FIG. **8**.

(127) For illustration of the operations shown in FIG. **6**, as shown in FIG. **10A**, the conductive rail

P02 corresponds to the first conductive rail, and the conductive segment **1052** corresponds to the first conductive segment that contacts the first conductive rail. With reference to the operation **S650** in FIG. **6** and FIG. **10A**, the dielectric structure **760** is formed and covers the conductive segment **1052**. The dielectric structure **760** also covers the conductive segment **1051**.

(128) Regarding the operations **S660-S670** in FIG. **6** and FIG. **10B**, part of structures are removed and are filled with the conductive material, to form a via **771** and the structure **781** which corresponds to a signal rail, as discussed with reference to FIGS. **6** and **7F-7G**. The via **771** and the structure **781** correspond to the via **832** and the signal rail **S11** respectively, as shown in FIG. **8**, in some embodiments. As shown in FIG. **10B**, the via **771** is formed on the **M0** layer and is connected to the structure **781** formed in the **M1** layer, which is similar to the embodiments illustrated in FIG. **7F**.

(129) In some embodiments, the conductive segment **1051** is referred to as the **M0** jumper, and corresponds to the conductive segment **821** shown in FIG. **8** or **9B**. In some other embodiments, the conductive segment **1052** is referred to as the **V0** rail, and corresponds to the conductive segment **812** shown in FIG. **8** or **9B**. In various embodiments, the via **771** corresponds to the via **433** shown in FIG. **8** or **9C**.

(130) In some embodiments, the method further includes the following operations. The dielectric structure is further patterned to form other patterns other than that are formed in the operation **S630**. These patterns are subsequently filled with the conductive material, to form a via that contacts the first conductive segment formed in the operation **S640**. Such via contacts the first conductive segment and a power rail disposed next to the second conductive rail. For illustration in FIG. **10C**, in some embodiments, the first conductive segment corresponds to the conductive segment **1052**, and the power rail corresponds to the structure **782** which corresponds to a power rail, thereby, the via corresponds to the via **1072** that contacts the first conductive segment and the power rail.

(131) As shown in FIG. **10C**, a via **1072** is formed on the conductive segment **1052** and is connected to the structure **782** formed in the **M1** layer. One terminal of the via **1072** is formed on the conductive segment **1052**, and the other terminal of the via **1072** is formed in the **M1** layer. With the operations discussed above, a height of via **1072** is smaller than a distance between the conductive rail **P02** and the structure **782**. Also, a height of via **1072** is smaller than a height of via **771** shown in at least one of FIG. **7F**, **7G** or **10B**.

(132) In some embodiments, the via **1072** is configured to couple between the conductive segment **1052** and the structure **782** formed in the **M1** layer. In various embodiments, the structure **782** is referred to as the power rail, for providing power signals. The conductive segment **1052** is referred to as the **V0** rail, and corresponds to the conductive segment **811** shown in FIG. **9A**. Also, the via **1072** corresponds to the via **831** shown in FIG. **9A**. Accordingly, the via **1072** is coupled between the conductive segment **1052** and the power rail **P11** shown in FIG. **8** or **9A**.

(133) The configuration of FIGS. **10A**, **10B**, and **10C** is given for illustrative purposes. Various configurations of the elements mentioned above in FIGS. **10A**, **10B**, and **10C** are within the contemplated scope of the present disclosure.

(134) To implement various devices, the layout diagrams as discussed above with respect to FIGS. **1A**, **1B**, **2**, **3A** to **3C**, **4**, **5A** to **5C**, **8** and **9A** to **9C** are used or modified to be used, as illustrated by the non-limiting examples discussed below with respect to FIGS. **11A** to **19B**. In the various embodiments discussed below, the IC of the present disclosure is implemented through the use of layout diagrams depicted in FIGS. **11A**, **12A**, **13A**, **14A**, **15A**, **16A**, **17A**, **18A** and **19A** that correspond to circuit diagrams depicted in FIGS. **11B**, **12B**, **13B**, **14B**, **15B**, **16B**, **17B** to **17E**, **18B** and **19B**, as indicated. To indicate correspondence between a given layout diagram feature formed based on the given layout diagram feature, a same reference designator is used in each of the layout diagram and structure depictions, as discussed below.

(135) Reference is now made to FIG. **11A**. FIG. **11A** is a circuit diagram of an IC **1100A**, in

accordance with some embodiments of the present disclosure. For illustration of the IC **1100A**, a gate terminal of a p-type metal oxide semiconductor (PMOS) transistor **P1** is coupled to a gate terminal of an n-type metal oxide semiconductor (NMOS) transistor **N1**. A source/drain terminal of the PMOS transistor **P1** is coupled to a source/drain terminal of a PMOS transistor **P2** at a node **A1**. A source/drain terminal of the NMOS transistor **N1** is coupled to a source/drain terminal of an NMOS transistor **N2** at a node **B1**. The node **A1** is further coupled to the node **B1** as indicated by connection **ZA** shown in FIG. **11A**. To implement the IC **1100A** including the connection **ZA** between the nodes **A1** and **B1** in the embodiments of the present disclosure, embodiments of layout designs and/or structures are provided and discussed below as illustrated with reference to FIG. **11B**.

(136) In some embodiments, the IC **1100A** is used as a unit cell or unit circuit, in which the unit cell or unit circuit is capable of being used as a basic unit or as part of a device or circuit, in order to implement various devices or circuits. Alternatively stated, in some embodiments, the IC **1100A** is implemented in various devices or circuits, including, for example, an inverter, a NAND gate, an AND-OR-invert (AOI) logic gate, a flip-flop, or the like.

(137) FIG. **11B** is a layout diagram **1100B** of an IC corresponding to the IC **1100A** of FIG. **11A**, in accordance with some embodiments of the present disclosure. For simplicity of illustration of the layout diagram **1100B**, it is merely illustrates a part of layout diagram for implement the IC **1100A** of FIG. **11A**.

(138) In the illustration of FIG. **11B**, gates **1111**, **1112** and **1113** are arranged in an active area (not shown) and extend along the column direction. In some embodiments, the gates **1111**, **1112** and **1113** correspond to the gate segments **311** and **312** shown in FIGS. **3A** to **3C**. In various embodiments, the gates **1111**, **1112** and **1113** are patterned as “Poly” illustrated in FIG. **11B**. Conductive segments **1121**, **1122**, **1123**, **1124** and **1125** extend along the column direction and are referred to as MD segments hereinafter. In various embodiments, the conductive segments **1121**, **1122**, **1123**, **1124** and **1125** are patterned as “MD” illustrated in FIG. **11B**.

(139) The MD segments **1121** and **1122** are arranged in the active area, as sources/drains of the PMOS transistor **P1** and **P2**. The gate **1112** and the MD segments **1121** and **1122** together correspond to the PMOS transistor **P1**. The gate **1113**, the MD segment **1122** and another MD segment (not shown) arranged next to the MD segment **1122** together correspond to the PMOS transistor **P2**. In such embodiments, the PMOS transistors **P1** and **P2** share the MD segment **1122**, which corresponds to the PMOS transistors **P1** and **P2** being coupled at the node **A1** as discussed above with respect to FIG. **11A**. The MD segments **1123**, **1124** and **1125** are arranged in the active area, as sources/drains of the NMOS transistor **N1** and **N2**. The gate **1111** and the MD segments **1123** and **1124** together correspond to the NMOS transistor **N1**. The gate **1112** and the MD segments **1124** and **1125** together correspond to the NMOS transistor **N2**. In such embodiments, the NMOS transistors **N1** and **N2** share the MD segment **1124**, which corresponds to the NMOS transistors **N1** and **N2** being coupled at the node **B1** as discussed above with respect to FIG. **11A**.

(140) Vias **1131** and **1132** are arranged between the active area and the **M0** layer. In various embodiments, the vias **1131** and **1132** are patterned as “VD” illustrated in FIG. **11B**. The vias **1131** and **1132** are also referred to as VD vias hereinafter, which are discussed above at least with reference to FIG. **5B**, for coupling between the MD segments **1121-1125** in the active area and conductive rails **1141-1143** in the **M0** layer. In the illustration of FIG. **11B**, the VD via **1131** is arranged between the MD segment **1122** and the conductive rail **1141**, and the VD via **1132** is arranged between the MD segment **1124** and the conductive rail **1142**. The VD via **1131** couples the MD segment **1122** to the conductive rail **1141**. The VD via **1132** couples the MD segment **1124** to the conductive rail **1142**.

(141) In some embodiments, the layout diagram **1100B** further includes a data rail (not shown) arranged in the **M1** layer above the **M0** layer, and the data rail extends in the column direction and is coupled to the conductive rail **1141**. With such configuration, a data signal provided from the

data rail is transmitted through the conductive rail **1141**, and is further transmitted through the via **1131** to the MD segment **1122**. In such embodiments, the node **A1** as discussed above with respect to FIG. **11A** is further coupled to the data rail, for receiving the data signal at the node **A1**.

(142) The Conductive rails **1141**, **1142** and **1143** are arranged in the **M0** layer which is above the active area and extend along the row direction. In some embodiments, the conductive rails **1141**, **1142** and **1143** are patterned as “**M0**” illustrated in FIG. **11B**. In some other embodiments, the conductive rails **1141**, **1142** and **1143** are also referred to as signal conductive rails which are discussed above at least with reference to FIG. **4**.

(143) A conductive segment **1151** is arranged in the **M0** layer, and is arranged below the **M1** layer. In various embodiments, the conductive segment **1151** is patterned as “**M01**” illustrated in FIG. **11B**. The conductive segment **1151** is shaped as a block and is also referred to as the **M0** jumper corresponding to the conductive segments **121-124** as illustrated in FIGS. **3A-3C**, in some embodiments. In a layout view, the conductive segment **1151** is partially overlapped with the conductive rails **1141** and **1142**, the gate **1111** and the VD via **1132**. The conductive segment **1151** couples the conductive rails **1141** and **1142** together. Accordingly, the MD segment **1122** is coupled through the VD via **1131** to the conductive rail **1141**, and the conductive rail **1141** is further coupled through the conductive segment **1151** to the conductive rail **1142**. The conductive rail **1142** is coupled through the VD via **1132** to the MD segment **1124**. With the above configurations, the MD segments **1122** and **1124** together are coupled to the conductive segment **1151**, which corresponds to the nodes **A1** and **B1** being coupled between the connection **ZA** as discussed above with respect to FIG. **11A**.

(144) Reference is now made to FIG. **12A**. FIG. **12A** is a circuit diagram of an IC **1200A**, in accordance with some embodiments of the present disclosure. For illustration of the IC **1200A**, a gate terminal of a PMOS transistor **P1** is coupled to a gate terminal of a NMOS transistor **N1**; a gate terminal of a PMOS transistor **P2** is coupled to a gate terminal of a NMOS transistor **N2**; a gate terminal of a PMOS transistor **P3** is coupled to a gate terminal of a NMOS transistor **N3**; and a gate terminal of a PMOS transistor **P4** is coupled to a gate terminal of a NMOS transistor **N4**.

(145) A source/drain terminal of the PMOS transistor **P1** is coupled to a node **A1**; a source/drain terminal of the PMOS transistor **P1** is coupled to a source/drain terminal of the PMOS transistor **P2** at a node **A2**; a source/drain terminal of the PMOS transistor **P2** is coupled to a source/drain terminal of the PMOS transistor **P3** at a node **A3**; a source/drain terminal of the PMOS transistor **P3** is coupled to a source/drain terminal of the PMOS transistor **P4** at a node **A4**; and a source/drain terminal of the PMOS transistor **P4** is coupled to a node **A5** which is further coupled to the nodes **A1** and **A3**.

(146) Furthermore, a source/drain terminal of the NMOS transistor **N1** is coupled to a node **B1**; a source/drain terminal of the NMOS transistor **N1** is coupled to a source/drain terminal of a NMOS transistor **N2**; a source/drain terminal of the NMOS transistor **N2** is coupled to a source/drain terminal of a NMOS transistor **N3** at a node **B2**; a source/drain terminal of the NMOS transistor **N3** is coupled to a source/drain terminal of a NMOS transistor **N4**; and a source/drain terminal of the NMOS transistor **N4** is coupled to a node **B3**. The node **A3** is further coupled to a power rail referenced as **VDD**. Each of the nodes **B1** and **B3** is further coupled to another power rail referenced as **VSS**, and a voltage of the power rail **VSS** is lower than that of the power rail **VDD**. The node **A4** is further coupled to the node **B2** as indicated by connection **ZN** shown in FIG. **12A**. To implement the IC **1200A** including the connection **ZN** between the nodes **A4** and **B2** in the embodiments of the present disclosure, embodiments of layout designs and/or structures are provided and discussed below as illustrated with reference to FIG. **12B**.

(147) FIG. **12B** is a layout diagram **1200B** of an IC corresponding to the IC **1200A** of FIG. **12A**, in accordance with some embodiments of the present disclosure.

(148) In the illustration of FIG. **12B**, gates **1211**, **1212**, **1213** and **1214** are arranged as gate terminals of PMOS transistors **P1-P4** or NMOS transistors **N1-N4** in FIG. **12A**. MD segments

1220, 1221, 1222, 1223, 1224, 1225, 1226, 1227, 1228 and 1229 are arranged as sources/drains of PMOS transistors **P1-P4** or NMOS transistors **N1-N4** in FIG. **12A**.

(149) The gate **1211** and the MD segments **1220** and **1221** together correspond to the PMOS transistor **P1**. The gate **1212** and the MD segments **1221** and **1222** together correspond to the PMOS transistor **P2**. The gate **1213** and the MD segments **1222** and **1223** together correspond to the PMOS transistor **P3**. The gate **1214** and the MD segments **1223** and **1224** together correspond to the PMOS transistor **P4**. In such embodiments, the PMOS transistors **P1** and **P2** share the MD segment **1221**, which corresponds to the PMOS transistors **P1** and **P2** being coupled at the node **A2** illustrated in FIG. **12A**. The PMOS transistors **P2** and **P3** share the MD segment **1222**, which corresponds to the PMOS transistors **P2** and **P3** being coupled at the node **A3** illustrated in FIG. **12A**. The PMOS transistors **P3** and **P4** share the MD segment **1223**, which corresponds to the PMOS transistors **P3** and **P4** being coupled at the node **A4** illustrated in FIG. **12A**.

(150) Furthermore, the gate **1211** and the MD segments **1225** and **1226** together correspond to the NMOS transistor **N1**. The gate **1212** and the MD segments **1226** and **1227** together correspond to the NMOS transistor **N2**. The gate **1213** and the MD segments **1227** and **1228** together correspond to the NMOS transistor **N3**. The gate **1214** and the MD segments **1228** and **1229** together correspond to the NMOS transistor **N4**. In such embodiments, the NMOS transistors **N2** and **N3** share the MD segment **1227**, which corresponds to the NMOS transistors **N2** and **N3** being coupled at the node **B2** illustrated in FIG. **12A**.

(151) Conductive rails **1241, 1242, 1243, 1244, 1245, 1246, 1247, 1248** and **1249** are arranged. The conductive rails **1242, 1243, 1244, 1245, 1246, 1248** and **1249** are arranged between the conductive rails **1241** and **1247**. A width of the conductive rails **1242, 1243, 1244, 1245, 1246, 1248** or **1249** is smaller than a width of the conductive rails **1241** or **1247**. In some embodiments, the conductive rails **1242, 1243, 1244, 1245, 1246, 1248** and **1249** are referred to as the signal conductive rails, and the conductive rails **1241** and **1247** are referred to as the power conductive rails, which are discussed above at least with reference to FIG. **4**.

(152) VD vias **1231, 1232, 1233, 1234, 1235, 1236, 1237** and **1238** are arranged. The VD via **1231** couples the MD segment **1220** to the conductive rail **1242**. The VD via **1232** couples the MD segment **1221** to the conductive rail **1241** which is further coupled to a power rail (not shown) arranged in the **M1** layer. The MD segment **1221** and the conductive rail **1241** together are coupled to the power rail, which corresponds to the node **A3** being coupled to the power rail **VDD** as discussed above with respect to FIG. **12A**. The VD via **1233** couples the MD segment **1222** to the conductive rail **1242**. The VD via **1234** couples the MD segment **1223** to the conductive rail **1244**. The VD via **1235** couples the MD segment **1224** to the conductive rail **1242**. With such configurations, the MD segments **1220, 1222** and **1224** are coupled together, which corresponds to the nodes **A1, A3** and **A5** being coupled together as discussed above with respect to FIG. **12A**.

(153) Moreover, the VD via **1236** couples the MD segment **1225** to the conductive rail **1247** which is further coupled to another power rail (not shown) arranged in the **M1** layer. The MD segment **1225** and the conductive rail **1247** together are coupled to the power rail, which corresponds to the node **B1** being coupled to the power rail **VSS** as discussed above with respect to FIG. **12A**. The VD via **1237** couples the MD segment **1227** to the conductive rail **1245**. The VD via **1238** couples the MD segment **1229** to the conductive rail **1247** which is further coupled to the power rail as same as that is coupled to the MD segment **1225**. The MD segment **1229** and the conductive rail **1247** together are coupled to the power rail, which corresponds to the node **B3** being coupled to the power rail **VSS** as discussed above with respect to FIG. **12A**.

(154) VG vias **1251, 1252, 1253** and **1254** are arranged. The VG via **1251** couples the gate **1211** to the conductive rail **1243**. The VG via **1252** couples the gate **1212** to the conductive rail **1246**. The VG via **1253** couples the gate **1213** to the conductive rail **1249**. The VG via **1254** couples the gate **1214** to the conductive rail **1248**.

(155) Conductive segments **1261, 1262** and **1263** are arranged. The conductive segment **1261** is

also referred to as the **M0** jumper, and corresponds to the conductive segment **121**, **122** or **123** as illustrated in FIG. 3A or 3C, in some embodiments. The conductive segment **1261** couples the conductive rails **1244** and **1245** together. With such configurations, the MD segments **1223** and **1227** are coupled together, which corresponds to the nodes **A4** and **B2** being coupled between the connection **ZN** as discussed above with respect to FIG. 12A.

(156) Moreover, the conductive segments **1262** and **1263** are also referred to as the **V0** rails, and correspond to the conductive segments **111** and **112** as illustrated in FIGS. 3A-3C, in some embodiments. As discussed above with reference to the VD vias **1231-1238**, the conductive segment **1262** couples to both of the conductive rail **1241** and the MD segment **1221**, which is further coupled to the power rail **VDD**. The conductive segment **1263** couples to both of the conductive rail **1247** and the MD segments **1225** and **1229**, which is further coupled to the power rail **VSS**.

(157) Reference is now made to FIG. 13A. FIG. 13A is a circuit diagram of an IC **1300A**, in accordance with some embodiments of the present disclosure. In some embodiments, the IC **1300A** is used as one unit cell/circuit for implementing two different logic functions, which is also indicated as a two stage cell. For illustration of the IC **1300A**, a gate terminal of a PMOS transistor **P1** is coupled to a gate terminal of a NMOS transistor **N1** at a node **E1**; a gate terminal of a PMOS transistor **P2** is coupled to a gate terminal of a NMOS transistor **N2**; and a gate terminal of a PMOS transistor **P3** is coupled to a gate terminal of a NMOS transistor **N3**.

(158) Furthermore, a source/drain terminal of the PMOS transistor **P1** is coupled to a node **A1**; a source/drain terminal of the PMOS transistor **P1** is coupled to a source/drain terminal of a PMOS transistor **P2** at a node **A2**; a source/drain terminal of the PMOS transistor **P2** is coupled to a source/drain terminal of a PMOS transistor **P3** at a node **A3**; a source/drain terminal of the PMOS transistor **P3** is coupled to a node **A4**. A source/drain terminal of the NMOS transistor **N1** is coupled to a node **B1**; a source/drain terminal of the NMOS transistor **N1** is coupled to a source/drain terminal of a NMOS transistor **N2** at a node **B2**; a source/drain terminal of the NMOS transistor **N2** is coupled to a source/drain terminal of a NMOS transistor **N3**; and a source/drain terminal of the NMOS transistor **N3** is coupled to a node **B3**. The nodes **A2** and **A4** are further coupled to a power rail referenced as **VDD**. The nodes **B2** is further coupled to another power rail referenced as **VSS**. With reference to FIG. 13A, the node **A1** is further coupled to the node **B1** as indicated by connection **Z**. The node **A3** is further coupled to the node **E1** at a node **E2** as indicated by connection **ZE**. The node **A3** is also further coupled to the node **B3** as indicated by connection **ZN**. To implement the IC **1300A** including the connection **Z** between the nodes **A1** and **B1**, the connection **ZE** between the nodes **E1** and **E2**, and connection **ZN** between the nodes **A3** and **B3** in the embodiments of the present disclosure, embodiments of layout designs and/or structures are provided and discussed below as illustrated with reference to FIG. 13B.

(159) FIG. 13B is a layout diagram **1300B** of an IC corresponding to the IC **1300A** of FIG. 13A, in accordance with some embodiments of the present disclosure.

(160) In the illustration of FIG. 13B, gates **1311**, **1312** and **1313** are arranged as gate terminals of PMOS transistors **P1-P3** or NMOS transistors **N1-N3** in FIG. 13A. MD segments **1321**, **1322**, **1323**, **1324**, **1325**, **1326** and **1327** are arranged as sources/drains of PMOS transistors **P1-P3** or NMOS transistors **N1-N3** in FIG. 13A.

(161) The gate **1311** and the MD segments **1321** and **1322** together correspond to the PMOS transistor **P1**. The gate **1312** and the MD segments **1322** and **1333** together correspond to the PMOS transistor **P2**. The gate **1313** and the MD segments **1323** and **1324** together correspond to the PMOS transistor **P3**. In such embodiments, the PMOS transistors **P1** and **P2** share the MD segment **1322**, which corresponds to the PMOS transistors **P1** and **P2** being coupled at the node **A2** illustrated in FIG. 13A. The PMOS transistors **P2** and **P3** share the MD segment **1323**, which corresponds to the PMOS transistors **P2** and **P3** being coupled at the node **A3** illustrated in FIG. 13A.

(162) Furthermore, the gate **1311** and the MD segments **1321** and **1325** together correspond to the NMOS transistor N1. The gate **1312** and the MD segments **1325** and **1326** together correspond to the NMOS transistor N2. The gate **1313** and the MD segments **1326** and **1327** together correspond to the NMOS transistor N3. In such embodiments, the NMOS transistors N1 and N2 share the MD segment **1325**, which corresponds to the NMOS transistors N1 and N2 being coupled at the node B2 illustrated in FIG. 13A. The PMOS transistor P1 and the NMOS transistor N1 share the MD segment **1321**, which corresponds to the PMOS transistor P1 and the NMOS transistor N1 coupled at the nodes A1 and B1 together. It also corresponds to the nodes A1 and B1 being coupled between the connection Z illustrated in FIG. 13A.

(163) Conductive rails **1341**, **1342**, **1343**, **1344**, **1345**, **1346** and **1347** are arranged. In some embodiments, the conductive rails **1342**, **1343**, **1344**, **1345** and **1346** are referred to as the signal conductive rails, and the conductive rails **1341** and **1347** are referred to as the power conductive rails, which are discussed above at least with reference to FIG. 4.

(164) VD vias **1331**, **1332**, **1333**, **1334**, **1335** and **1336** are arranged. The VD via **1331** couples the MD segment **1321** to the conductive rail **1344**. The VD via **1332** couples the MD segment **1322** to the conductive rail **1341** which is further coupled to a power rail (not shown) arranged in the M1 layer. The MD segment **1322** and the conductive rail **1341** together are coupled to the power rail, which corresponds to the node A2 being coupled to the power rail VDD as discussed above with respect to FIG. 13A. The VD via **1333** couples the MD segment **1323** to the conductive rail **1343**. The VD via **1334** couples the MD segment **1324** to the conductive rail **1341**. Similarly, the MD segment **1324** and the conductive rail **1341** together are coupled to the power rail, which corresponds to the node A4 being coupled to the power rail VDD as discussed above with respect to FIG. 13A. The VD via **1335** couples the MD segment **1325** to the conductive rail **1347** which is further coupled to another power rail (not shown) arranged in the M1 layer. The MD segment **1325** and the conductive rail **1347** together are coupled to the power rail, which corresponds to the node B2 being coupled to the power rail VSS as discussed above with respect to FIG. 13A. The VD via **1336** couples the MD segment **1327** to the conductive rail **1346**.

(165) VG vias **1351**, **1352** and **1353** are arranged. The VG via **1351** couples the gate **1311** to the conductive rail **1343**. With such configurations, the gate **1311** is further coupled through the conductive rail **1343** to the MD segment **1323**, which corresponds to the nodes E1 and A3 being coupled between the connection ZE as discussed above with respect to FIG. 13A. The VG via **1352** couples the gate **1312** to the conductive rail **1345**. The VG via **1353** couples the gate **1313** to the conductive rail **1342**.

(166) Conductive segments **1361**, **1362** and **1363** are arranged. The conductive segment **1361** is also referred to as the M0 jumper, and corresponds to the conductive segment **121**, **122** or **123** as illustrated in FIG. 3A or 3C, in some embodiments. The conductive segment **1361** couples the conductive rails **1343** and **1346** together. With such configurations, the MD segments **1323** and **1327** are coupled together, which corresponds to the nodes A3 and B3 being coupled between the connection ZN as discussed above with respect to FIG. 13A.

(167) Moreover, the conductive segments **1362** and **1363** are also referred to as the V0 rails, and correspond to the conductive segments **111** and **112** as illustrated in FIGS. 3A-3C, in some embodiments. As discussed above with reference to the VD vias **1331-1336**, the conductive segment **1362** couples to both of the conductive rail **1341** and the MD segments **1322** and **1324**, which is further coupled to the power rail VDD. The conductive segment **1263** couples to both of the conductive rail **1347** and the MD segment **1325**, which is further coupled to the power rail VSS.

(168) Reference is now made to FIG. 14A. FIG. 14A is a circuit diagram of an IC **1400A**, in accordance with some embodiments of the present disclosure. For illustration of the IC **1400A**, a gate terminal of a PMOS transistor P1 is coupled to a gate terminal of a NMOS transistor N1 as indicated by connection I. In some embodiments, the connection I is indicated as an input terminal,

for providing a control signal to both of the PMOS transistor P1 and the NMOS transistor N1. (169) Furthermore, a source/drain terminal of the PMOS transistor P1 is coupled to a node A1. A source/drain terminal of the PMOS transistor P1 is coupled to a node A2. A source/drain terminal of the NMOS transistor N1 is coupled to a node B1. A source/drain terminal of the NMOS transistor N2 is coupled to a node B2. The nodes A1 is further coupled to a power rail referenced as VDD. The nodes B1 is further coupled to another power rail referenced as VSS. The node A2 is further coupled to the node B2 as indicated by connection ZN. To implement the IC 1400A including the connection ZN between the nodes A2 and B2 in the embodiments of the present disclosure, embodiments of layout designs and/or structures are provided and discussed below as illustrated with reference to FIG. 14B.

(170) FIG. 14B is a layout diagram 1400B of an IC corresponding to the IC 1400A of FIG. 14A, in accordance with some embodiments of the present disclosure.

(171) In the illustration of FIG. 14B, a gate 1411 is arranged as the gate terminal of PMOS transistor P1 and NMOS transistor N1 in FIG. 14A. MD segments 1421, 1422 and 1423 are arranged as sources/drains of PMOS transistor P1 or NMOS transistor N1 in FIG. 14A.

(172) The gate 1411 and the MD segments 1421 and 1422 together correspond to the PMOS transistor P1. The gate 1411 and the MD segments 1423 and 1422 together correspond to the NMOS transistor N1. In such embodiments, the PMOS transistor P1 share the MD segment 1422, which corresponds to the PMOS transistor P1 being coupled at the nodes A2 and B2 together illustrated in FIG. 14A. It also corresponds to the nodes A2 and B2 being coupled between the connection ZN illustrated in FIG. 14A.

(173) Conductive rails 1441, 1442, 1443, 1444, 1445 and 1446 are arranged. In some embodiments, the conductive rails 1442, 1443, 1444 and 1445 are referred to as the signal conductive rails, and the conductive rails 1441 and 1446 are referred to as the power conductive rails, which are discussed above at least with reference to FIG. 4.

(174) VD vias 1431, 1432 and 1433 are arranged. The VD via 1431 couples the MD segment 1421 to the conductive rail 1441 which is further coupled to a power rail (not shown) arranged in the M1 layer. The MD segment 1421 and the conductive rail 1441 together are coupled to the power rail, which corresponds to the node A1 being coupled to the power rail VDD as discussed above with respect to FIG. 14A. The VD via 1432 couples the MD segment 1422 to the conductive rail 1444. The VD via 1433 couples the MD segment 1423 to the conductive rail 1446 which is further coupled to another power rail (not shown) arranged in the M1 layer. The MD segment 1423 and the conductive rail 1446 together are coupled to the power rail, which corresponds to the node B1 being coupled to the power rail VSS as discussed above with respect to FIG. 14A.

(175) VG via 1451 is arranged. The VG via 1451 couples the gate 1411 to the conductive rail 1445 which is further coupled to a signal rail (not shown) arranged in the M1 layer. The gate 1411 and the conductive rail 1445 together are coupled to the signal rail, which corresponds to the gate of the PMOS transistor P1 or NMOS transistor N1 being coupled between the connection I as discussed above with respect to FIG. 14A.

(176) Conductive segments 1461, 1462 and 1463 are arranged. The conductive segment 1461 is also referred to as the M0 jumper, and corresponds to the conductive segment 121, 122 or 123 as illustrated in FIG. 3A or 3C, in some embodiments. The conductive segment 1461 couples the conductive rails 1443 and 1444 together.

(177) Moreover, the conductive segments 1462 and 1463 are also referred to as the V0 rails, and correspond to the conductive segments 111 and 112 as illustrated in FIGS. 3A-3C, in some embodiments. As discussed above with reference to the VD vias 1431-1433, the conductive segment 1462 couples to both of the conductive rail 1441 and the MD segment 1421, which is further coupled to the power rail VDD. The conductive segment 1463 couples to both of the conductive rail 1446 and the MD segment 1423, which is further coupled to the power rail VSS.

(178) Reference is now made to FIG. 15A. FIG. 15A is a circuit diagram of an IC 1500A, in

accordance with some embodiments of the present disclosure. For illustration of the IC **1500A**, a gate terminal of a PMOS transistor **P1** is coupled to a gate terminal of a NMOS transistor **N1** as indicated by connection **I**. A gate terminal of a PMOS transistor **P2** is coupled to a gate terminal of a NMOS transistor **N2** as also indicated by the connection **I**. In some embodiments, the connection **I** has a similar configuration/arrangement to the connection **I** illustrated in FIG. **14A**.

(179) Furthermore, a source/drain terminal of the PMOS transistor **P1** is coupled to a node **A1**; a source/drain terminal of the PMOS transistor **P1** is coupled to a source/drain terminal of a PMOS transistor **P2** at a node **A2**; and a source/drain terminal of the PMOS transistor **P2** is coupled to a node **A3**. A source/drain terminal of the NMOS transistor **N1** is coupled to a node **B1**; a source/drain terminal of the NMOS transistor **N1** is coupled to a source/drain terminal of a NMOS transistor **N2** at node **B2**; and a source/drain terminal of the NMOS transistor **N2** is coupled to a node **B3**. The nodes **A1** and **A3** are further coupled to a power rail referenced as **VDD**. The nodes **B1** and **B3** are further coupled to another power rail referenced as **VSS**. The node **A2** is further coupled to the node **B2** as indicated by connection **ZN** shown in FIG. **15A**. To implement the IC **1500A** including the connection **ZN** between the nodes **A2** and **B2** in the embodiments of the present disclosure, embodiments of layout designs and/or structures are provided and discussed below as illustrated with reference to FIG. **15B**.

(180) FIG. **15B** is a layout diagram **1500B** of an IC corresponding to the IC **1500A** of FIG. **15A**, in accordance with some embodiments of the present disclosure.

(181) In the illustration of FIG. **15B**, gates **1511** and **1512** are arranged as gate terminals of PMOS transistors **P1-P2** or NMOS transistors **N1-N2** in FIG. **15A**. MD segments **1521**, **1522**, **1523**, **1524** and **1525** are arranged as source/drain terminals of PMOS transistors **P1-P2** or NMOS transistors **N1-N2** in FIG. **15A**.

(182) The gate **1511** and the MD segments **1521** and **1522** together correspond to the PMOS transistor **P1**. The gate **1512** and the MD segments **1522** and **1523** together correspond to the PMOS transistor **P2**. In such embodiments, the PMOS transistors **P1** and **P2** share the MD segment **1522**, which corresponds to the PMOS transistors **P1** and **P2** being coupled at the node **A2** illustrated in FIG. **15A**. The gate **1511** and the MD segments **1524** and **1522** together correspond to the NMOS transistor **N1**. The gate **1512** and the MD segments **1522** and **1525** together correspond to the NMOS transistor **N2**. In such embodiments, the NMOS transistors **N1** and **N2** share the MD segment **1522**, which corresponds to the NMOS transistors **N1** and **N2** being coupled at the node **B2** illustrated in FIG. **15A**. Also, the NMOS transistor **N1/N2** and the PMOS transistor **P1/P2** share the MD segment **1522**, which corresponds to the nodes **A2** and **B2** being coupled between the connection **ZN** illustrated in FIG. **15A**.

(183) Conductive rails **1541**, **1542**, **1543**, **1544**, **1545** and **1546** are arranged. In some embodiments, the conductive rails **1542**, **1543**, **1544** and **1545** are referred to as the signal conductive rails, and the conductive rails **1541** and **1546** are referred to as the power conductive rails, which are discussed above at least with reference to FIG. **4**.

(184) VD vias **1531**, **1532**, **1533**, **1534** and **1535** are arranged. The VD via **1531** couples the MD segment **1521** to the conductive rail **1541** which is further coupled to a power rail (not shown) arranged in the **M1** layer. The MD segment **1521** and the conductive rail **1541** together are coupled to the power rail, which corresponds to the node **A1** being coupled to the power rail **VDD** as discussed above with respect to FIG. **15A**. The VD via **1532** couples the MD segment **1522** to the conductive rail **1543**. The VD via **1533** couples the MD segment **1523** to the conductive rail **1541**. Similarly, the MD segment **1523** and the conductive rail **1541** together are coupled to the power rail, which corresponds to the node **A3** being coupled to the power rail **VDD** as discussed above with respect to FIG. **15A**.

(185) Moreover, the VD via **1534** couples the MD segment **1524** to the conductive rail **1546** which is further coupled to another power rail (not shown) arranged in the **M1** layer. The MD segment **1524** and the conductive rail **1546** together are coupled to the power rail, which corresponds to the

node B1 being coupled to the power rail VSS as discussed above with respect to FIG. 15A. The VD via 1535 couples the MD segment 1525 to the conductive rail 1546. Similarly, the MD segment 1525 and the conductive rail 1546 together are coupled to the power rail, which corresponds to the node B3 being coupled to the power rail VSS as discussed above with respect to FIG. 15A.

(186) VG vias 1551 and 1552 are arranged. The VG via 1551 couples the gate 1511 to the conductive rail 1545 which is further coupled to a signal rail (not shown) arranged in the M1 layer. Also, the VG via 1552 couples the gate 1512 to the conductive rail 1545. The gate 1511 and the conductive rail 1545 together are coupled to the signal rail, which corresponds to the gate of the PMOS transistor P1 or NMOS transistor N1 being coupled between the connection I as discussed above with respect to FIG. 15A. Similarly, the gate 1512 and the conductive rail 1545 together are coupled to the same signal rail that is also coupled to the gate 1511, which corresponds to the gate of the PMOS transistor P2 or NMOS transistor N2 being coupled between the connection that is also indicated as the connection I.

(187) Conductive segments 1561, 1562 and 1563 are arranged. The conductive segment 1561 is also referred to as the M0 jumper, and corresponds to the conductive segment 124 as illustrated in FIG. 3B or 3C, in some embodiments. The conductive segment 1561 couples the conductive rails 1542 and 1543 together.

(188) Moreover, the conductive segments 1562 and 1563 are also referred to as the V0 rails, and correspond to the conductive segments 111 and 112 as illustrated in FIGS. 3A-3C, in some embodiments. As discussed above with reference to the VD vias 1531-1535, the conductive segment 1562 couples to both of the conductive rail 1541 and the MD segments 1521 and 1523, which is further coupled to the power rail VDD. The conductive segment 1563 couples to both of the conductive rail 1546 and the MD segments 1524 and 1525, which is further coupled to the power rail VSS.

(189) Reference is now made to FIG. 16A. FIG. 16A is a circuit diagram of an IC 1600A, in accordance with some embodiments of the present disclosure. For illustration of the IC 1600A, a gate terminal of a PMOS transistor P1 is coupled to a gate terminal of a NMOS transistor N1 as indicated by connection I1. A gate terminal of a PMOS transistor P2 is coupled to a gate terminal of a NMOS transistor N2 as indicated by connection I2; and a gate terminal of a PMOS transistor P3 is coupled to a gate terminal of a NMOS transistor N3 as indicated by connection I3. In some embodiments, the connection I1, I2 or I3 has a similar configuration/arrangement to the connection I illustrated in FIG. 14A. In some other embodiments, the connections I1, I2 and I3 are further coupled to signal rails (not shown) that are different from one another, for transmitting different data signals. For example, the connection I1 is coupled to a first signal rail; the connection I2 is coupled to a second signal rail; and the connection I3 is coupled to a third signal rail.

(190) Furthermore, a source/drain terminal of the PMOS transistor P1 is coupled to a node A1; a source/drain terminal of the PMOS transistor P1 is coupled to a source/drain terminal of a PMOS transistor P2 at a node A2; a source/drain terminal of the PMOS transistor P2 is coupled to a source/drain terminal of a PMOS transistor P3 at a node A3. A source/drain terminal of the PMOS transistor P3 is coupled to a node A4. A source/drain terminal of the NMOS transistor N1 is coupled to a node B1; a source/drain terminal of the NMOS transistor N1 is coupled to a source/drain terminal of a NMOS transistor N2; a source/drain terminal of the NMOS transistor N2 is coupled to a source/drain terminal of a NMOS transistor N3; and a source/drain terminal of the NMOS transistor N3 is coupled to a node B2. The nodes A1 and A3 are further coupled to a power rail referenced as VDD. The node B2 is further coupled to another power rail referenced as VSS. The node A2 is further coupled to the node A4. The node A2 is further coupled to the node B1 as indicated by connection ZN shown in FIG. 16A. To implement the IC 1600A including the connection ZN between the nodes A2 and B1 in the embodiments of the present disclosure, embodiments of layout designs and/or structures are provided and discussed below as illustrated with reference to FIG. 16B.

(191) FIG. 16B is a layout diagram 1600B of an IC corresponding to the IC 1600A of FIG. 16A, in accordance with some embodiments of the present disclosure.

(192) In the illustration of FIG. 16B, gates 1611, 1612 and 1613 are arranged as gate terminals of PMOS transistors P1-P3 or NMOS transistors N1-N3 in FIG. 16A. MD segments 1621, 1622, 1623, 1624, 1625, 1626, 1627 and 1628 are arranged as source/drain terminals of PMOS transistors P1-P3 or NMOS transistors N1-N3 in FIG. 16A.

(193) The gate 1611 and the MD segments 1621 and 1622 together correspond to the PMOS transistor P1. The gate 1612 and the MD segments 1622 and 1623 together correspond to the PMOS transistor P2. In such embodiments, the PMOS transistors P1 and P2 share the MD segment 1622, which corresponds to the PMOS transistors P1 and P2 being coupled at the node A2 illustrated in FIG. 16A. The gate 1613 and the MD segments 1623 and 1624 together correspond to the PMOS transistor P3. In such embodiments, the PMOS transistors P2 and P3 share the MD segment 1623, which corresponds to the PMOS transistors P2 and P3 being coupled at the node A3 illustrated in FIG. 16A. The gate 1611 and the MD segments 1625 and 1626 together correspond to the NMOS transistor N1. The gate 1612 and the MD segments 1626 and 1627 together correspond to the NMOS transistor N2. The gate 1613 and the MD segments 1627 and 1628 together correspond to the NMOS transistor N3.

(194) Conductive rails 1641, 1642, 1643, 1644, 1645, 1646 and 1647 are arranged. In some embodiments, the conductive rails 1642, 1643, 1644, 1645 and 1646 are referred to as the signal conductive rails, and the conductive rails 1641 and 1647 are referred to as the power conductive rails, which are discussed above at least with reference to FIG. 4.

(195) VD vias 1631, 1632, 1633, 1634, 1635 and 1636 are arranged. The VD via 1631 couples the MD segment 1621 to the conductive rail 1641 which is further coupled to a power rail (not shown) arranged in the M1 layer. The MD segment 1621 and the conductive rail 1641 together are coupled to the power rail, which corresponds to the node A1 being coupled to the power rail VDD as discussed above with respect to FIG. 16A. The VD via 1632 couples the MD segment 1622 to the conductive rail 1643. The VD via 1633 couples the MD segment 1623 to the conductive rail 1641 which is further coupled to the power rail. Similarly, the MD segment 1623 and the conductive rail 1641 together are coupled to the power rail, which corresponds to the node A3 being coupled to the power rail VDD as discussed above with respect to FIG. 16A. The VD via 1634 couples the MD segment 1624 to the conductive rail 1643. With such configurations, the MD segments 1622, and 1624 are coupled together, which corresponds to the nodes A2 and A4 being coupled together as discussed above with respect to FIG. 16A.

(196) Furthermore, the VD via 1635 couples the MD segment 1625 to the conductive rail 1644. The VD via 1636 couples the MD segment 1628 to the conductive rail 1647 which is further coupled to another power rail (not shown) arranged in the M1 layer. The MD segment 1628 and the conductive rail 1647 together are coupled to the power rail, which corresponds to the node B2 being coupled to the power rail VSS as discussed above with respect to FIG. 16A.

(197) VG vias 1651, 1652 and 1653 are arranged. The VG via 1651 couples the gate 1611 to the conductive rail 1645 which is further coupled to a first signal rail (not shown) arranged in the M1 layer. The gate 1611 and the conductive rail 1645 together are coupled to the first signal rail, which corresponds to the connection I1 being coupled to the first signal rail as discussed above with respect to FIG. 16A, in some embodiments. The VG via 1652 couples the gate 1612 to the conductive rail 1642 which is further coupled to a second signal rail (not shown) arranged in the M1 layer. The gate 1612 and the conductive rail 1642 together are coupled to the second signal rail, which corresponds to the connection I2 being coupled to the second signal rail as discussed above with respect to FIG. 16A, in some embodiments. The VG via 1653 couples the gate 1613 to the conductive rail 1646 which is further coupled to a third signal rail (not shown) arranged in the M1 layer. The gate 1613 and the conductive rail 1646 together are coupled to the third signal rail, which corresponds to the connection I3 being coupled to the third signal rail as discussed above

with respect to FIG. 16A, in some embodiments.

(198) Conductive segments **1661**, **1662** and **1663** are arranged. The conductive segment **1661** is shaped as a railed block, and is also referred to as the **M0** jumper, corresponding to the conductive segment **124** as illustrated in FIG. 3B or 3C, in some embodiments. The conductive segment **1661** couples the conductive rails **1643** and **1644** together. With such configurations, the MD segments **1625** and **1622** are coupled together, which corresponds to the nodes B1 and A2 being coupled between the connection ZN as discussed above with respect to FIG. 16A. In addition, the MD segments **1625**, **1622** and **1624** are further coupled together, which corresponds to the nodes B1, A2 and A4 being coupled to each other as discussed above with respect to FIG. 16A.

(199) Moreover, the conductive segments **1662** and **1663** are also referred to as the **V0** rails, and correspond to the conductive segments **111** and **112** as illustrated in FIGS. 3A-3C, in some embodiments. As discussed above with reference to the VD vias **1631-1636**, the conductive segment **1662** couples to both of the conductive rail **1641** and the MD segments **1621** and **1623**, which is further coupled to the power rail VDD. The conductive segment **1663** couples to both of the conductive rail **1647** and the MD segment **1628**, which is further coupled to the power rail VSS.

(200) Reference is now made to FIG. 17A. FIG. 17A is a circuit diagram of an IC **1700A**, in accordance with some embodiments of the present disclosure. For illustration of the IC **1700A**, a PMOS transistor P1 includes a gate terminal A1, a drain (D) terminal and a source (S) terminal. A NMOS transistor N1 includes a gate terminal B2, a D terminal and an S terminal. The S terminal of the PMOS transistor P1 is coupled to the D terminal of the NMOS transistor N1 to form a first transistor pair. A PMOS transistor P2 includes a gate terminal B1, a D terminal and an S terminal. A NMOS transistor N2 includes a gate terminal A2, a D terminal and an S terminal. The D terminal of the PMOS transistor P2 is coupled to the S terminal of the NMOS transistor N2 to form a second transistor pair.

(201) The gate terminal A1 of the PMOS transistor P1 is coupled to the gate terminal A2 of the NMOS transistor N2. The gate terminal B1 of the PMOS transistor P2 is coupled to the gate terminal B2 of the NMOS transistor N1. The S terminal of the PMOS transistor P1 and the D terminal of the NMOS transistor N1 are coupled to the D terminal of the PMOS transistor P2 and the S terminal of the NMOS transistor N2, to indicate a connection Z as shown in FIG. 17A, in order to operate as a transmission gate circuit **1700A**. In some embodiments, the connection Z is further coupled to a signal rail and operated as an output terminal of the circuit **1700A**, for transmitting a data signal. In some embodiments, the transmission gate circuit **1700A** is in a single cell. To implement the transmission gate circuit **1700A** including the connection Z in the embodiments of the present disclosure, embodiments of layout designs and/or structures are provided as discussed in more detail below as illustrated with reference to FIG. 17B, 17C or 17D.

(202) Reference is now made to FIGS. 17B and 17C. FIGS. 17B and 17C are layout diagrams **1700B** and **1700C** of ICs corresponding to the IC **1700A** of FIG. 17A, in accordance with some embodiments of the present disclosure. For simplicity of illustration of the layout diagram **1700B** or **1700C**, it is merely illustrates a part of layout diagram for implement the IC **1700A** of FIG. 17A.

(203) In some embodiments, the layout diagram **1700B** or **1700C** illustrates a unit pattern, in which the unit pattern is capable of being fabricated as a basic unit or as part of a device or circuit including the transmission gate as discussed above with respect to FIG. 17A, in order to implement various layout diagrams. Alternatively stated, in some embodiments, the layout diagram **1700B** or **1700C** is implemented in various layout diagrams for implementing the devices or circuits including the transmission gate.

(204) As illustrated in FIG. 17B, the layout diagram **1700B** includes gates **1711** and **1712**, MD segments (not labelled), conductive rails **1741**, **1742** and **1743**, VG vias **1751** and **1752**, a conductive segment **1761**, and cut segments **1771** and **1772**.

(205) The cut segments **1771** and **1772** arranged in the active area and extend along the row

direction, patterned as “CT” in FIG. 17B, for cutting off conductive segments in the layout diagram 1700B. For example, the cut segment 1771 is configured to cut off the gate 1711, and the cut segment 1772 is configured to cut off the gate 1712. Alternatively stated, the cut segments 1771 and 1772 are configured to separate one gate 1711 or 1712 to two separated portions. In some embodiments, the cut segments 1771 and 1772 correspond to the cut segments CT at least shown in FIG. 1A.

(206) The VG via 1751 couples the gate 1711 with the VG via 1751 to the conductive rail 1743, and VG via 1752 couples the gate 1712 with the VG via 1752 to the conductive rail 1741.

(207) The conductive segment 1761 is partially overlapped with the gate 1711, the cut segment 1771, the conductive rails 1741 and 1743, and the VG via 1751. The conductive segment 1761 is also referred to as the M0 jumper, and corresponds to the conductive segment 121, 122 or 123 as illustrated in FIG. 3A or 3C, in some embodiments. The conductive segment 1761 couples the conductive rails 1741 and 1743 together.

(208) In the illustration of FIG. 17B, a poly pitch D1 is referred to as a distance between two adjacent gates 1711 and 1712. A M0 pitch D2 is referred to as a distance between two adjacent conductive rails 1741 and 1742. In some embodiments, the poly pitch D1 corresponds to the poly pitch D1 shown in FIGS. 3A to 3C, and the M0 pitch D2 corresponds to the M0 pitch D2 shown in FIGS. 3A to 3C.

(209) The cut segments 1771 and 1772 are separated by a distance D3 in the column direction. The distance D3 is referred to as a jog interval between two adjacent edges of these two adjacent cut segments 1771 and 1772, as illustrated in FIG. 17B. In addition, the gate 1711 is separated from the cut segment 1772 by a distance D4 in the row direction. The cut segment 1772 is not arranged across the gate 1711 and is arranged next to the gate 1711. The distance D4 is referred to as an interval between an edge of the cut segment 1772 and an edge of the gate 1711. The edge of the cut segment 1772 and the edge of the gate 1711 are arranged next to each other.

(210) In some embodiments, a minimum of the distance D3 is about in a range of 0.3 times of the M0 pitch D2 to 0.6 times of the M0 pitch D2 (i.e., $0.3 \times \text{M0 pitch D2} \sim 1 \times \text{M0 pitch D2}$). In various embodiments, the distance D4 is about in a range of 0.4 times of the poly pitch D1 to 0.6 times of the poly pitch D1 (i.e., $0.4 \times \text{poly pitch D1} \sim 0.6 \times \text{poly pitch D1}$).

(211) As illustrated in FIG. 17C, the layout diagram 1700C includes gates 1713, 1714 and 1715, MD segments (not labelled), conductive rails 1744 and 1745, VG vias 1753 and 1754, a conductive segment 1762, and cut segments 1773 and 1774.

(212) The cut segment 1773 is partially overlapped with the gate 1713 and is configured to cut off the gate 1713. The cut segment 1774 is partially overlapped with the gate 1715 and is configured to cut off the gate 1715. In some embodiments, the cut segments 1773 and 1774 correspond to the cut segments CT at least shown in FIG. 1A.

(213) The VG via 1753 is overlapped with the gate 1713 and the conductive rail 1745, for coupling the gate 1713 with the VG via 1753 to the conductive rail 1745. The VG via 1754 is overlapped with the gate 1715 and the conductive rail 1744, for coupling the gate 1715 with the VG via 1754 to the conductive rail 1744.

(214) The conductive segment 1762 is partially overlapped with the gate 1715, the cut segment 1774, the conductive rails 1744 and 1745, and the VG via 1754. The conductive segment 1762 is also referred to as the M0 jumper, and corresponds to the conductive segment 121, 122 or 123 as illustrated in FIG. 3A or 3C, in some embodiments. The conductive segment 1762 couples the conductive rails 1744 and 1745 together.

(215) In the illustration of FIG. 17C, the poly pitch D1 and the M0 pitch D2 correspond to the poly pitch D1 and the M0 pitch D2 shown in FIG. 17B. The cut segments 1773 and 1774 are separated by a distance D5 in the column direction. The distance D5 is referred to as a jog interval between two adjacent edges of these two adjacent cut segments 1773 and 1774, as illustrated in FIG. 17C. The gate 1714 is separated from the cut segment 1774 by a distance D6 in the row direction. The

cut segment **1775** is not arranged across the gate **1714** and is arranged next to the gate **1714**. The distance **D6** is referred to as an interval between two adjacent edges of the cut segment **1774** and the gate **1714**.

(216) In some embodiments, the distance **D5** corresponds to the distance **D3** shown in FIG. **17B**, and has a similar range scope. In some other embodiments, the distance **D6** corresponds to the distance **D4** shown in FIG. **17B**, and has a similar range scope.

(217) Reference is now made to FIG. **17D**. FIG. **17D** is a layout diagram **1700D** of ICs corresponding to the IC **1700A** of FIG. **17A**, in accordance with some embodiments of the present disclosure.

(218) In the illustration of FIG. **17D**, gates **1711** and **1712** are arranged as gate terminals of PMOS transistors **P1-P2** or NMOS transistors **N1-N2** in FIG. **17A**. MD segments **1721**, **1722**, **1723**, **1724** and **1725** are arranged as source/drain terminals of PMOS transistors **P1-P2** or NMOS transistors **N1-N2** in FIG. **17A**.

(219) Cut segments **1771** and **1772** are arranged and will be removed for fabricating the layout diagram **1700D**. The cut segment **1771** is arranged across the gate **1711** for separating the gate **1711** to two portions including the gate **1711** with a VG via **1751** and the gate **1711** with a VG via **1753**. The cut segment **1771** is partially overlapped with a conductive rail **1743** and the gate **1711**. The cut segment **1772** is arranged across the gate **1712** for separating the gate **1712** to two portions including the gate **1712** with a VG via **1752** and the gate **1712** with a VG via **1754**. The cut segment **1772** is partially overlapped with a conductive rail **1744** and the gate **1712**.

(220) The gate **1711** with the VG via **1751** and the MD segments **1721** and **1722** together correspond to the PMOS transistor **P1**. The gate **1712** with the VG via **1752** and the MD segments **1722** and **1723** together correspond to the PMOS transistor **P2**. The gate **1711** with the VG via **1753** and the MD segments **1724** and **1722** together correspond to the NMOS transistor **N1**. The gate **1712** with the VG via **1754** and the MD segments **1722** and **1725** together correspond to the NMOS transistor **N2**. In such embodiments, the PMOS transistors **P1** and **P2** and the NMOS transistors **N1** and **N2** share the MD segment **1722**, which corresponds to the PMOS transistors **P1** and **P2** and the NMOS transistors **N1** and **N2** being coupled between the connection **Z** illustrated in FIG. **17A**.

(221) Conductive rails **1741**, **1742**, **1743**, **1744**, **1745** and **1746** are arranged. In some embodiments, the conductive rails **1741**, **1742**, **1743**, **1744**, **1745** and **1746** are referred to as the signal conductive rails, which are discussed above at least with reference to FIG. **4**.

(222) VD via **1731** is arranged. The VD via **1731** couples the MD segment **1722** to the conductive rail **1746** which is further coupled to a signal rail (not shown) arranged in the **M1** layer. In some embodiments, the signal rail is indicated as the output terminal for transmitting the signal transmitted from the connection **Z** as discussed above with respect to FIG. **17A**.

(223) VG vias **1751**, **1752**, **1753** and **1754** are arranged. The VG via **1751** couples the gate **1711** with the VG via **1751** to the conductive rail **1741** which is further coupled to a first signal rail (not shown) arranged in the **M1** layer. The VG via **1752** couples the gate **1712** with the VG via **1752** to the conductive rail **1743** which is further coupled to a second signal rail (not shown) arranged in the **M1** layer. The VG via **1753** couples the gate **1711** with the VG via **1753** to the conductive rail **1744** which is further coupled to the second signal rail. The VG via **1754** couples the gate **1712** with the VG via **1754** to the conductive rail **1745** which is further coupled to the first signal rail. In such configurations, it corresponds to the gate terminal **A1** of the PMOS transistor **P1** and the gate terminal **A2** of the NMOS transistor **N2** being coupled together to receive a first signal transmitted from the first signal rail, as discussed above with respect to FIG. **17A**. Similarly, it corresponds to the gate terminal **B1** of the PMOS transistor **P2** and the gate terminal **N2** of the NMOS transistor **B2** being coupled together to receive a second signal transmitted from the second signal rail, as discussed above with respect to FIG. **17A**.

(224) Conductive segment **1761** is arranged. The conductive segment **1761** is shaped as a block,

and is also referred to as the **M0** jumper, corresponding to the conductive segment **121**, **122** or **123** as illustrated in FIG. 3A or 3C, in some embodiments. The conductive segment **1761** is partially overlapped with the gate **1711**, the cut segment **1771**, two adjacent conductive rails **1743** and **1744**, and the VG via **1753**. The conductive segment **1761** couples the conductive rails **1743** and **1744** together. As discussed above, the conductive rails **1743** and **1744** are further coupled to the second signal rail by having the arrangement of the conductive segment **1761**.

(225) Reference is now made to FIG. 17E. FIG. 17E is a layout diagram **1700E** of ICs corresponding to the IC **1700A** of FIG. 17A, in accordance with some embodiments of the present disclosure.

(226) In the illustration of FIG. 17E, compared to the embodiments in FIG. 17D, the cut segment **1771** is partially overlapped with a conductive rail **1744** and the gate **1711**, and the cut segment **1772** is partially overlapped with a conductive rail **1745'** and the gate **1712**. The VG via **1753** couples the MD segment **1722** to the conductive rail **1743** which is further coupled to the signal rail (not shown) as an output rail for transmitting the signal transmitted from the connection Z as discussed above with respect to FIG. 17A. The VG via **1752** couples the gate **1712** with the VG via **1752** to the conductive rail **1744** which is further coupled to the second signal rail. The VG via **1753** couples the gate **1711** with the VG via **1753** to the conductive rail **1744** which is further coupled to the second signal rail. The VG via **1754** couples the gate **1712** with the VG via **1754** to the conductive rail **1745** which is further coupled to the first signal rail. In such configurations, it also corresponds to the connections illustrated in FIG. 17A, as discussed with reference FIG. 17D. The conductive segment **1761** is partially overlapped with the gate **1711**, the cut segment **1771**, two adjacent conductive rails **1744** and **1745**, and the VG via **1753**.

(227) Reference is now made to FIG. 18A. FIG. 18A is a circuit diagram of an IC **1800A** that is equivalent to the transmission gate circuit **1700A** in FIG. 17A, in accordance with some embodiments of the present disclosure. In the illustration of FIG. 18A, compared to FIG. 17A, the IC **1800A** further includes a PMOS transistor **P3** and a NMOS transistor **N3**. A gate terminal of the PMOS transistor **P1** is coupled to a node **A1**; a gate terminal of the PMOS transistor **P2** is coupled to a node **B1**; a gate terminal of the PMOS transistor **P3** is coupled to a gate of the NMOS transistor **N2**. A gate terminal of the NMOS transistor **N1** is coupled to a node **B2**; a gate terminal of the NMOS transistor **N3** is coupled to the gate of the PMOS transistor **P2**; the gate terminal of the NMOS transistor **N2** is coupled to a node **A2**. The node **A2** is further coupled to the gate terminal of the PMOS transistor **P3**. The node **B1** is further coupled to the node **B2** as indicated by connection Z shown in FIG. 18A. To implement the IC **1800A** including the connection Z between the nodes **B1** and **B2** in the embodiments of the present disclosure, embodiments of layout designs and/or structures are provided and discussed below as illustrated with reference to FIG. 18B.

(228) Furthermore, a source/drain terminal of the PMOS transistor **P1** is coupled to a source/drain terminal of the PMOS transistor **P2** at a node **S1**; a source/drain terminal of the PMOS transistor **P2** is coupled to a source/drain terminal of the PMOS transistor **P3** at a node **S2**; and a source/drain terminal of the PMOS transistor **P3** is coupled to a node **S2'**. A source/drain terminal of the NMOS transistor **N1** is coupled to a source/drain terminal of the NMOS transistor **N3** at a node **S3**; a source/drain terminal of the NMOS transistor **N3** is coupled to a source/drain terminal of the NMOS transistor **N2** at a node **S3'**. The node **S1** is further coupled to the node **S3** as indicated by connection Z' shown in FIG. 18A. To implement the IC **1800A** including the connection Z' between the nodes **S1** and **S3** in the embodiments of the present disclosure, embodiments of layout designs and/or structures are provided and discussed below as illustrated with reference to FIG. 18B.

(229) The node **S2** is further coupled to the node **S2'**, corresponding to the source terminal and the drain terminal of the PMOS transistor **P3** coupled together. The source and drain terminal of the PMOS transistor **P3** are coupled together in a short-circuit configuration such that the PMOS transistor **P3** is inoperative. In addition, the node **S3** is further coupled to the node **S3'**, corresponding to the source terminal and the drain terminal of the NMOS transistor **N3** coupled

together. The source and drain terminal of the NMOS transistor N3 are coupled together in a short-circuit configuration such that the NMOS transistor N3 is inoperative. With the connections of the PMOS transistor P3 and the NMOS transistor N3, as illustrated in FIG. 18A, the IC 1800A is able to operate as a circuit equivalent to the transmission gate circuit 1700A in FIG. 17A.

(230) FIG. 18B is a layout diagram 1800B of an IC corresponding to the IC 1800A of FIG. 18A, in accordance with some embodiments of the present disclosure.

(231) In the illustration of FIG. 18B, gates 1811, 1812 and 1813 are arranged as gate terminals of PMOS transistors P1-P3 or NMOS transistors N1-N3 in FIG. 18A. MD segments 1821, 1822, 1823, 1824, 1825, 1826 and 1827 are arranged as source/drain terminals of PMOS transistors P1-P3 or NMOS transistors N1-N3 in FIG. 18A.

(232) A cut segment 1871 is arranged and will be removed for fabricating the IC 1800A. The cut segment 1871 is arranged across the gate 1811 for separating the gate 1811 to two portions including the gate 1811 with a VG via 1851 and the gate 1811 with a VG via 1853.

(233) The gate 1811 with the VG via 1851 and the MD segments 1821 and 1822 together correspond to the PMOS transistor P1. The gate 1812 and the MD segments 1822 and 1823 together correspond to the PMOS transistor P2. In such embodiments, the PMOS transistors P1 and P2 share the MD segment 1822, which corresponds to the PMOS transistors P1 and P2 being coupled at the node S1 illustrated in FIG. 18A. The gate 1813 and the MD segments 1823 and 1824 together correspond to the PMOS transistor P3. In such embodiments, the PMOS transistors P2 and P3 share the MD segment 1823, which corresponds to the PMOS transistors P2 and P3 being coupled at the node S2 illustrated in FIG. 18A. The gate 1811 with the VG via 1853 and the MD segments 1825 and 1822 together correspond to the NMOS transistor N1. The gate 1812 and the MD segments 1822 and 1826 together correspond to the NMOS transistor N3. In such embodiments, the NMOS transistors N1 and N3 share the MD segment 1822, which corresponds to the NMOS transistors N1 and N3 being coupled at the node S3 illustrated in FIG. 18A.

Furthermore, the PMOS transistors P1 and P2 and the NMOS transistors N1 and N3 share the MD segment 1822, which corresponds to the PMOS transistors P1 and P2 and the NMOS transistors N1 and N3 being coupled between the connection Z' illustrated in FIG. 18A. The gate 1813 and the MD segments 1826 and 1827 together correspond to the NMOS transistor N2. In such embodiments, the NMOS transistors N3 and N2 share the MD segment 1826, which corresponds to the NMOS transistors N3 and N2 being coupled at the node S3' illustrated in FIG. 18A.

(234) Conductive rails 1841, 1842, 1843, 1844, 1845, 1846, 1847 and 1848 are arranged. In some embodiments, the conductive rails 1842, 1843, 1844, 1846, 1847 and 1848 are referred to as the signal conductive rails, and the conductive rails 1841 and 1845 are referred to as the power conductive rails, which are discussed above at least with reference to FIG. 4.

(235) VD vias 1831, 1832, 1833, 1834 and 1835 are arranged. The VD via 1831 couples the MD segment 1822 to the conductive rail 1842. The VD via 1832 couples the MD segment 1823 to the conductive rail 1846, and the VD via 1833 couples the MD segment 1824 to the conductive rail 1846. In such embodiments, the source terminal of the PMOS transistor P3 and the drain terminal of the PMOS transistor P3 are coupled together, which corresponds to the source/drain terminals of the PMOS transistor P3 being coupled at the nodes S2 and S2' as the short-circuit configuration illustrated in FIG. 18A. Similarly, the VD via 1834 couples the MD segment 1822 to the conductive rail 1845, and the VD via 1835 couples the MD segment 1826 to the conductive rail 1846. In such embodiments, the source terminal of the NMOS transistor N2 and the drain terminal of the NMOS transistor N2 are coupled together, which corresponds to the source/drain terminals of the NMOS transistor N2 being coupled together at the nodes S3 and S3' as the short-circuit configuration illustrated in FIG. 18A.

(236) VG vias 1851, 1852, 1853 and 1854 are arranged. The VG via 1851 couples the gate 1811 with the VG via 1851 to the conductive rail 1841 which is further coupled to a first signal rail (not shown) arranged in the M1 layer. It corresponds to the PMOS transistor P1 being coupled at the

node A1 in FIG. 18A. The VG via 1852 couples the gate 1812 to the conductive rail 1843. The VG via 1853 couples the gate 1812 with the VG via 1853 to the conductive rail 1844. The VG via 1854 couples the gate 1813 to the conductive rail 1847 which is further coupled to a second signal rail (not shown) arranged in the M1 layer. It corresponds to the NMOS transistor N2 being coupled at the node A2 in FIG. 18A.

(237) A conductive segment 1861 is arranged, and is partially overlapped with the cut segment 1871, the gate 1811, the conductive rails 1843 and 1844, and the VG via 1853. The conductive segment 1861 is shaped as a block, and is also referred to as a M0 jumper, corresponding to the conductive segment 121, 122 or 123 as illustrated in FIG. 3A or 3C, in some embodiments. The conductive segment 1861 couples the conductive rails 1843 and 1844 together. With such configurations, the gate 1811 with the VG via 1853 and the gate 1812 are coupled together, which corresponds to the nodes B1 and B2 being coupled between the connection Z as discussed above with respect to FIG. 18A.

(238) Reference is now made to FIG. 19A. FIG. 19A is a circuit diagram of an IC 1900A that is equivalent to the transmission gate circuit 1700A in FIG. 17A, in accordance with some embodiments of the present disclosure. In the illustration of FIG. 19A, compared to FIG. 17A, the IC 1900A further includes PMOS transistor P3, P4, P5 and P6 and NMOS transistors N3, N4, N5 and N6. A gate terminal of the PMOS transistor P1 is coupled to a gate terminal of the NMOS transistor N1 at a node C1; a gate terminal of the PMOS transistor P2 is coupled to a gate terminal of the NMOS transistor N2 as indicated by connection I1; a gate terminal of the PMOS transistor P3 is coupled to a node SB1; a gate terminal of the PMOS transistor P4 is coupled to a gate terminal of the NMOS transistor N4 as indicated by connection I2; a gate terminal of the PMOS transistor P5 is coupled to a node S1; a gate terminal of the PMOS transistor P6 is coupled to a gate terminal of the NMOS transistor N6 as indicated by connection S2; a gate terminal of the NMOS transistor N3 is coupled to a node S3; and a gate terminal of the NMOS transistor N5 is coupled to a node SB3.

(239) In some embodiments for illustrating the FIGS. 19A and 19B, the nodes SB1, SB2 and the connection SB3 are further coupled to a first signal rail indicated as SB (not shown), for receiving a first signal transmitted from the first signal rail. The nodes S1, S2 and S3 are further coupled to a second signal rail indicated as S (not shown), for receiving a second signal, other than the first signal, transmitted from the signal rail. The connection I1 is further coupled to a third signal rail indicated as I1 (not shown), for receiving a third signal transmitted from the third signal rail. The connection I2 is further coupled to a fourth signal rail indicated as I2 (not shown), for receiving a fourth signal transmitted from the fourth signal rail.

(240) Furthermore, a source/drain terminal of the PMOS transistor P1 is coupled to a node A1; a source/drain terminal of the PMOS transistor P1 is coupled to a source/drain terminal of a PMOS transistor P2 at a node A2; a source/drain terminal of the PMOS transistor P2 is coupled to a source/drain terminal of a PMOS transistor P3; a source/drain terminal of the PMOS transistor P3 is coupled to a source/drain terminal of a PMOS transistor P4 at a node A3; a source/drain terminal of the PMOS transistor P4 is coupled to a source/drain terminal of a PMOS transistor P5; a source/drain terminal of the PMOS transistor P5 is coupled to a source/drain terminal of a PMOS transistor P6 at a node A4; and a source/drain terminal of the PMOS transistor P6 is coupled to a node A5. A source/drain terminal of the NMOS transistor N1 is coupled to a node B1; a source/drain terminal of the NMOS transistor N1 is coupled to a source/drain terminal of a NMOS transistor N2 at a node B2; a source/drain terminal of the NMOS transistor N2 is coupled to a source/drain terminal of a NMOS transistor N3; a source/drain terminal of the NMOS transistor N3 is coupled to a source/drain terminal of a NMOS transistor N4 at a node B3; a source/drain terminal of the NMOS transistor N4 is coupled to a source/drain terminal of a NMOS transistor N5; a source/drain terminal of the NMOS transistor N5 is coupled to a source/drain terminal of a NMOS transistor N6 at a node B4; and a source/drain terminal of the NMOS transistor N6 is

coupled to a node B5.

(241) The nodes A2 and A4 are further coupled to a power rail referenced as VDD. The nodes B2 and B4 are further coupled to another power rail referenced as VSS. The node C2 is further coupled to the node C1. The node A1 is further coupled to the node B1 as indicated by connection Z. The node A3 is further coupled to the node B3 at a node C2. The node C2 is further coupled to the node C1 as indicated by connection S4. The node A5 is further coupled to the node B5 as indicated by connection SB2. To implement the IC 1900A including the connection Z between the nodes A1 and B1, the connection S4 between the nodes C1 and C2, and the connection SB2 between the nodes A5 and B5, in the embodiments of the present disclosure, embodiments of layout designs and/or structures are provided and discussed below as illustrated with reference to FIG. 19B.

(242) FIG. 19B is a layout diagram 1900B of an IC corresponding to the IC 1900A of FIG. 19A, in accordance with some embodiments of the present disclosure. Compared to the layout 1800B in FIG. 18B, less conductive rails are arranged in the layout diagram 1900B.

(243) In the illustration of FIG. 19B, gates 1911, 1912, 1913, 1914, 1915 and 1916 are arranged as gate terminals of PMOS transistors P1-P6 or NMOS transistors N1-N6 in FIG. 18A. MD segments 1920, 1920', 1921, 1922, 1923, 1924, 1925, 1926, 1927, 1928 and 1929 are arranged as source/drain terminals of PMOS transistors P1-P6 or NMOS transistors N1-N6 in FIG. 19A. Cut segments 1971 and 1972 are arranged. The cut segment 1971 is arranged across the gate 1913 for separating the gate 1913 to two portions including the gate 1913 with a VG via 1953 and the gate 1913 with a VG via 1954. The cut segment 1972 is arranged across the gate 1915 for separating the gate 1915 to two portions including the gate 1915 with a VG via 1956 and the gate 1915 with a VG via 1957.

(244) The gate 1911 and the MD segments 1920 and 1921 together correspond to the PMOS transistor P1. The gate 1912 and the MD segments 1921 and 1922 together correspond to the PMOS transistor P2. In such embodiments, the PMOS transistors P1 and P2 share the MD segment 1921, which corresponds to the PMOS transistors P1 and P2 being coupled at the node A2 illustrated in FIG. 19A. The gate 1913 with the VG via 1953 and the MD segments 1922 and 1923 together correspond to the PMOS transistor P3. The gate 1914 and the MD segments 1923 and 1924 together correspond to the PMOS transistor P4. In such embodiments, the PMOS transistors P3 and P4 share the MD segment 1923, which corresponds to the PMOS transistors P3 and P4 being coupled at the node A3 illustrated in FIG. 19A. The gate 1915 with the VG via 1956 and the MD segments 1924 and 1925 together correspond to the PMOS transistor P5. The gate 1916 and the MD segments 1925 and 1926 together correspond to the PMOS transistor P6. In such embodiments, the PMOS transistors P5 and P6 share the MD segment 1925, which corresponds to the PMOS transistors P5 and P6 being coupled at the node A4 illustrated in FIG. 19A.

(245) Furthermore, the gate 1911 and the MD segments 1920 and 1927 together correspond to the NMOS transistor N1. In such embodiments, the NMOS transistor N1 and the PMOS transistor P1 share the MD segment 1920, which corresponds to the NMOS transistor N1 and the PMOS transistor P1 being coupled between illustrated the connection Z in FIG. 19A. The gate 1912 and the MD segments 1927 and 1928 together correspond to the NMOS transistor N2. In such embodiments, the NMOS transistors N1 and N2 share the MD segment 1927, which corresponds to the NMOS transistors N1 and N2 being coupled at the node B2 in FIG. 19A. The gate 1913 and the MD segments 1928 and 1923 together correspond to the NMOS transistor N3. The gate 1914 and the MD segments 1923 and 1929 together correspond to the NMOS transistor N4. In such embodiments, the NMOS transistors N3-N4 and the PMOS transistors P3-P4 share the MD segment 1923, which corresponds to the NMOS transistors N3-N4 and the PMOS transistors P3-P4 being coupled between the nodes A3 and B3 in FIG. 19A. The gate 1915 with the VG via 1957 and the MD segments 1929 and 1920' together correspond to the NMOS transistor N5. The gate 1916 and the MD segments 1920' and 1926 together correspond to the NMOS transistor N6. In such embodiments, the NMOS transistors N5 and N6 share the MD segment 1920', which corresponds

to the NMOS transistors N5 and N6 being coupled at the node B4 in FIG. 19A. In addition, the NMOS transistor N6 and the PMOS transistor P6 share the MD segment 1926, which corresponds to the NMOS transistor N6 and the PMOS transistor P6 being coupled between the nodes A5 and B5 in FIG. 19A.

(246) Conductive rails 1940, 1940', 1941, 1942, 1943, 1944, 1945, 1946, 1947, 1948 and 1949 are arranged. In some embodiments, the conductive rails 1940, 1940', 1942, 1943, 1944, 1945, 1947, 1948 and 1949 are referred to as signal conductive rails, and the conductive rails 1941 and 1946 are referred to as power conductive rails, which are discussed above at least with reference to FIG. 4.

(247) VD vias 1931, 1932, 1933, 1934, 1935, 1936 and 1937 are arranged. The VD via 1931 couples the MD segment 1920 to the conductive rail 1942. The VD via 1932 couples the MD segment 1921 to the conductive rail 1941 which is further coupled to a power rail (not shown) arranged in the M1 layer. The MD segment 1921 and the conductive rail 1941 together are coupled to the power rail, which corresponds to the node A2 being coupled to the power rail VDD as discussed above with respect to FIG. 19A. The VD via 1933 couples the MD segment 1923 to the conductive rail 1944. The VD via 1934 couples the MD segment 1925 to the conductive rail 1941. The MD segment 1925 and the conductive rail 1941 together are coupled to the power rail, which corresponds to the node A4 being coupled to the power rail VDD as discussed above with respect to FIG. 19A.

(248) Moreover, the VD via 1935 couples the MD segment 1927 to the conductive rail 1946 which is further coupled to another power rail (not shown) arranged in the M1 layer. The MD segment 1927 and the conductive rail 1946 together are coupled to such power rail, which corresponds to the node B2 being coupled to the power rail VD VSS as discussed above with respect to FIG. 19A. The VD via 1936 couples the MD segment 1920' to the conductive rail 1946 which is further coupled to the power rail (not shown) that is same as the conductive rail 1946 is coupled to. The MD segment 1920' and the conductive rail 1946 together are coupled to such power rail, which corresponds to the node B4 being coupled to the power rail VD VSS as discussed above with respect to FIG. 19A. The VD via 1937 couples the MD segment 1926 to the conductive rail 1940 which is further coupled to the first signal rail (not shown) arranged in the M1 layer.

(249) VG vias 1951, 1952, 1953, 1954, 1955, 1957, 1958 and 1959 are arranged. The VG via 1951 couples the gate 1911 to the conductive rail 1944 which is further coupled to a signal rail (not shown) arranged in the M1 layer. It corresponds to the PMOS transistor P1 and the NMOS transistor N1 being coupled at the node C11 in FIG. 19A. In such configurations, the MD segment 1923 coupled to the conductive rail 1944 is further coupled to the gate 1911. It corresponds to the nodes C1 and C2 which is also coupled to the nodes A3 and B3 being coupled between the connection S4 in FIG. 19A. The VG via 1952 couples the gate 1912 to the conductive rail 1945 which is further coupled to the third signal rail (not shown) arranged in the M1 layer. It corresponds to the PMOS transistor P2 and the NMOS transistor N2 being between the connection I1 in FIG. 19A. The VG via 1953 couples the gate 1913 with the VG via 1953 to the conductive rail 1949 which is further coupled to the first signal rail. It corresponds to the PMOS transistor P3 being coupled at the node SB1 in FIG. 19A. The VG via 1954 couples the gate 1913 with the VG via 1954 to the conductive rail 1940' which is further coupled to the second signal rail (not shown) arranged in the M1 layer, in some embodiments. It corresponds to the NMOS transistor N3 being coupled at the node S3 in FIG. 19A.

(250) Moreover, the VG via 1955 couples the gate 1914 to the conductive rail 1947 which is further coupled to the fourth signal rail (not shown) arranged in the M1 layer, in some embodiments. It corresponds to the PMOS transistor P4 and the NMOS transistor N4 being between the connection I2 in FIG. 19A. The VG via 1956 couples the gate 1915 with the VG via 1956 to the conductive rail 1948 which is further coupled to the second signal rail. It corresponds to the PMOS transistor P5 being coupled at the node S1 in FIG. 19A. The VG via 1957 couples the gate 1915 with the VG via 1957 to the conductive rail 1940 which is further coupled to the first signal rail. It corresponds

to the NMOS transistor N5 being coupled at the node SB3 in FIG. 19A. The VG via 1958 couples the gate 1916 to the conductive rail 1948 which is further coupled to the second signal rail. Also, the VG via 1959 couples the gate 1916 to the conductive rail 1940' which is further coupled to the second signal rail as well. It corresponds to the PMOS transistor P6 and the NMOS transistor N6 being between the connection S2 in FIG. 19A.

(251) Conductive segments 1961, 1962, 1963 and 1964 are arranged. The conductive segment 1961 is shaped as a railed block, and is also referred to as a M0 jumper, corresponding to the conductive segment 124 as illustrated in FIG. 3B or 3C, in some embodiments. The conductive segment 1961 couples the conductive rails 1942 and 1943 together.

(252) Moreover, the conductive segment 1962 is shaped as a block, and is also referred to as a M0 jumper, corresponding to the conductive segment 121, 122 or 123 as illustrated in FIG. 3A or 3C, in some embodiments. The conductive segment 1962 is partially overlapped with the gate 1915, the cut segment 1972, the conductive rails 1949 and 1940, and the VG via 1957. The conductive segment 1962 couples the conductive rails 1949 and 1940 together. With such configurations, the gate 1913, the gate 1915 with the VG via 1957 and the MD segment 1926 are coupled together, which corresponds to the PMOS transistor P3, the NMOS transistor N5, and the nodes B1 and A2 coupled between the connection SB2 being coupled to a same signal rail which, in some embodiments, is the first signal rail, as discussed above with respect to FIG. 19A.

(253) Furthermore, the conductive segments 1963 and 1964 are also referred to as the V0 rails, and correspond to the conductive segments 111 and 112 as illustrated in FIGS. 3A-3C, in some embodiments. As discussed above with reference to the VD vias 1931-1937, the conductive segment 1963 couples to both of the conductive rail 1941 and the MD segments 1921 and 1925, which is further coupled to the power rail VDD. The conductive segment 1964 couples to both of the conductive rail 1946 and the MD segments 1927 and 1920', which is further coupled to the power rail VSS.

(254) As illustrated by the non-limiting examples depicted in FIGS. 1A to 19B and discussed above, the various embodiments include conductive segments formed on conductive rails and disposed above the M0 layer and below the M1 layer, thereby increasing routing flexibility compared to approaches in which lack of these conductive segments.

(255) Reference is now made to FIG. 20. FIG. 20 is a flow chart of a method 2000 for fabricating an IC, in accordance with some embodiments of the present disclosure. In some embodiments, the IC is manufactured based on an IC layout diagram includes one of layout diagrams 100A, 100B, 200, 300A-300C, 400, 500A-500C, 800, 1100B, 1200B, 1300B, 1400B, 1500B, 1600B, 1700B-1700E, 1800B, or 1900B, corresponding to an IC structure, e.g., one of ICs 700, or 1000, discussed above with respect to FIGS. 1A-19B. In some embodiments, the operations of method 2000 are performed in the order depicted in FIG. 20. In some embodiments, the operations of method 2000 are performed simultaneously and/or in an order other than the order depicted in FIG. 20. In some embodiments, one or more operations are performed before, between, during, and/or after performing one or more operations of method 2000.

(256) At operation 2010, conductive rails are formed in a first metal layer. In some embodiments, the conductive rails correspond to the conductive rails P01, P02 and S01-S04 discussed above with respect to FIGS. 1A to 19B. The first metal layer correspond to the M0 layer discussed above with respect to FIGS. 1A to 19B.

(257) In some embodiments, the conductive rails include a pair of first conductive rails. In various embodiments, the pair of first conductive rails correspond to the conductive rails P01 and P02, that are also indicated as power conductive rails, discussed above with respect to FIGS. 1A to 19B.

(258) At operation 2020, signal rails are formed in a second metal layer above the first metal layer. In some embodiments, the signal rails correspond to the signal rails P11, P12 and S11-S4 or rails not shown discussed above with respect to FIGS. 1A to 19B. The second metal layer correspond to the M1 layer discussed above with respect to FIGS. 1A to 19B.

(259) At operation **2030**, a first conductive segment is formed between the first metal layer and the second metal layer, and is spaced apart from the signal rails. In some embodiments, the first conductive segment is overlapped with the pair of first conductive rails in a layout view. In some embodiments, the first conductive segment corresponds to the conductive segment indicated as the **V0** rail discussed above with respect to FIGS. **1A** to **19B**. For example, the first conductive segment corresponds to the conductive segment **111** or **112** shown in FIGS. **1A**, **1B**, **3A** to **3C**, or the conductive segment **411a**, **411b**, **412a**, **412b** in FIGS. **4**, **5A** to **5C**, etc.

(260) In some embodiments, the method **2000** further includes the operations as follows. A second conductive segment is also formed between the first metal layer and the second metal layer, and is also spaced apart from the signal rails. The second conductive segment is overlapped with at least two adjacent conductive rails between the pair of first conductive rail in a layout view. In some embodiments, the second conductive segment corresponds to the conductive segment indicated as the **M0** jumper discussed above with respect to FIGS. **1A** to **19B**. For example, the second conductive segment corresponds to the conductive segment **121**, **122**, **123** or **124** shown in FIGS. **1A**, **1B**, **3A** to **3C**, or the conductive segment **421** in FIGS. **4**, **5A** to **5C**, etc. In various embodiments, the adjacent conductive rails disposed with the second conductive segment corresponds to the conductive rails **S0**, that are also indicated as signal conductive rails, discussed above with respect to FIGS. **1A** to **19B**.

(261) Reference is now made to FIG. **21**. FIG. **21** is a block diagram of an electronic design automation (EDA) system **2100** for designing the integrated circuit layout design, in accordance with some embodiments of the present disclosure. EDA system **2100** is configured to implement one or more operations of the method **600** disclosed in FIG. **6** and the method **2000** disclosed in FIG. **20**, and further explained in conjunction with FIGS. **1A-5C** and **7A-19B**. In some embodiments, EDA system **2100** includes an APR system.

(262) In some embodiments, EDA system **2100** is a general purpose computing device including a hardware processor **2120** and a non-transitory, computer-readable storage medium **2160**. Storage medium **2160**, amongst other things, is encoded with, i.e., stores, computer program code (instructions) **2161**, i.e., a set of executable instructions. Execution of instructions **2161** by hardware processor **2120** represents (at least in part) an EDA tool which implements a portion or all of, e.g., the method **600** or **2000**.

(263) The processor **2120** is electrically coupled to computer-readable storage medium **2160** via a bus **2150**. The processor **2120** is also electrically coupled to an I/O interface **2110** and a fabrication tool **2170** by bus **2150**. A network interface **2130** is also electrically connected to processor **2120** via bus **2150**. Network interface **2130** is connected to a network **2140**, so that processor **2120** and computer-readable storage medium **2160** are capable of connecting to external elements via network **2140**. The processor **2120** is configured to execute computer program code **2161** encoded in computer-readable storage medium **2160** in order to cause EDA system **2100** to be usable for performing a portion or all of the noted processes and/or methods. In one or more embodiments, processor **2120** is a central processing unit (CPU), a multi-processor, a distributed processing system, an application specific integrated circuit (ASIC), and/or a suitable processing unit.

(264) In one or more embodiments, computer-readable storage medium **2160** is an electronic, magnetic, optical, electromagnetic, infrared, and/or a semiconductor system (or apparatus or device). For example, computer-readable storage medium **2160** includes a semiconductor or solid-state memory, a magnetic tape, a removable computer diskette, a random access memory (RAM), a read-only memory (ROM), a rigid magnetic disk, and/or an optical disk. In one or more embodiments using optical disks, computer-readable storage medium **2160** includes a compact disk-read only memory (CD-ROM), a compact disk-read/write (CD-R/W), and/or a digital video disc (DVD).

(265) In one or more embodiments, storage medium **2160** stores computer program code **2161** configured to cause EDA system **2100** (where such execution represents (at least in part) the EDA

tool) to be usable for performing a portion or all of the noted processes and/or methods. In one or more embodiments, storage medium **2160** also stores information which facilitates performing a portion or all of the noted processes and/or methods. In one or more embodiments, storage medium **2160** stores library **2162** of standard cells including such standard cells as disclosed herein, for example, a cell including conductive segments **111**, **112** or **121-123** discussed above with respect to FIG. **1A**.

(266) EDA system **2100** includes I/O interface **2110**. I/O interface **2110** is coupled to external circuitry. In one or more embodiments, I/O interface **2110** includes a keyboard, keypad, mouse, trackball, trackpad, touchscreen, and/or cursor direction keys for communicating information and commands to processor **2120**.

(267) EDA system **2100** also includes network interface **2130** coupled to processor **2120**. Network interface **2130** allows EDA system **2100** to communicate with network **2140**, to which one or more other computer systems are connected. Network interface **2130** includes wireless network interfaces such as BLUETOOTH, WIFI, WIMAX, GPRS, or WCDMA; or wired network interfaces such as ETHERNET, USB, or IEEE-1364. In one or more embodiments, a portion or all of noted processes and/or methods, is implemented in two or more systems **2100**.

(268) EDA system **2100** also includes the fabrication tool **2170** coupled to the processor **2120**. The fabrication tool **2170** is configured to fabricate integrated circuits, including, for example, the integrated circuit **700** illustrated in FIGS. **7A-7G** or the integrated circuit **1000** illustrated in FIGS. **10A-10C**, based on the design files processed by the processor **2120** and/or the IC layout designs as discussed above.

(269) EDA system **2100** is configured to receive information through I/O interface **2110**. The information received through I/O interface **2110** includes one or more of instructions, data, design rules, libraries of standard cells, and/or other parameters for processing by processor **2120**. The information is transferred to processor **2120** via bus **2150**. EDA system **2100** is configured to receive information related to a UI through I/O interface **2110**. The information is stored in computer-readable medium **2160** as user interface (UI) **2163**.

(270) In some embodiments, a portion or all of the noted processes and/or methods is implemented as a standalone software application for execution by a processor. In some embodiments, a portion or all of the noted processes and/or methods is implemented as a software application that is a part of an additional software application. In some embodiments, a portion or all of the noted processes and/or methods is implemented as a plug-in to a software application. In some embodiments, at least one of the noted processes and/or methods is implemented as a software application that is a portion of an EDA tool. In some embodiments, a portion or all of the noted processes and/or methods is implemented as a software application that is used by EDA system **2100**. In some embodiments, a layout diagram which includes standard cells is generated using a tool such as VIRTUOSO® available from CADENCE DESIGN SYSTEMS, Inc., or another suitable layout generating tool.

(271) In some embodiments, the processes are realized as functions of a program stored in a non-transitory computer readable recording medium. Examples of a non-transitory computer readable recording medium include, but are not limited to, external/removable and/or internal/built-in storage or memory unit, for example, one or more of an optical disk, such as a DVD, a magnetic disk, such as a hard disk, a semiconductor memory, such as a ROM, a RAM, a memory card, and the like.

(272) FIG. **22** is a block diagram of IC manufacturing system **2200**, and an IC manufacturing flow associated therewith, in accordance with some embodiments. In some embodiments, based on a layout diagram, at least one of (A) one or more semiconductor masks or (B) at least one component in a layer of a semiconductor integrated circuit is fabricated using IC manufacturing system **2200**.

(273) In FIG. **22**, IC manufacturing system **2200** includes entities, such as a design house **2210**, a mask house **2220**, and an IC manufacturer/fabricator ("fab") **2230**, that interact with one another in

the design, development, and manufacturing cycles and/or services related to manufacturing an IC device **2240**. The entities in IC manufacturing system **2200** are connected by a communications network. In some embodiments, the communications network is a single network. In some embodiments, the communications network is a variety of different networks, such as an intranet and the Internet. The communications network includes wired and/or wireless communication channels. Each entity interacts with one or more of the other entities and provides services to and/or receives services from one or more of the other entities. In some embodiments, two or more of design house **2210**, mask house **2220**, and IC fab **2230** is owned by a single larger company. In some embodiments, two or more of design house **2210**, mask house **2220**, and IC fab **2230** coexist in a common facility and use common resources.

(274) Design house (or design team) **2210** generates an IC design layout diagram **2211**. IC design layout diagram **2211** includes various geometrical patterns, for example, an IC layout design depicted in FIGS. **1A-1B**, FIG. **2**, FIG. **3A-3C**, FIG. **4**, FIGS. **5A-5C**, FIG. **8**, FIGS. **9A-9C**, FIG. **11B**, FIG. **12B**, FIG. **13B**, FIG. **14B**, FIG. **15B**, FIG. **16B**, FIG. **17B**, FIG. **18B**, and/or FIG. **19B**, designed for an IC device **2240**, for example, integrated circuits **700** and **1000**, discussed above with respect to FIGS. **7A-7G** and/or FIGS. **10A-10C**. The geometrical patterns correspond to patterns of metal, oxide, or semiconductor layers that make up the various components of IC device **2240** to be fabricated. The various layers combine to form various IC features. For example, a portion of IC design layout diagram **2211** includes various IC features, such as an active area, gate electrode, source and drain, conductive segments or vias of an interlayer interconnection, to be formed in a semiconductor substrate (such as a silicon wafer) and various material layers disposed on the semiconductor substrate. Design house **2210** implements a proper design procedure to form IC design layout diagram **2211**. The design procedure includes one or more of logic design, physical design or place and route. IC design layout diagram **2211** is presented in one or more data files having information of the geometrical patterns. For example, IC design layout diagram **2211** can be expressed in a GDSII file format or DFII file format.

(275) Mask house **2220** includes data preparation **2221** and mask fabrication **2222**. Mask house **2220** uses IC design layout diagram **2211** to manufacture one or more masks **2223** to be used for fabricating the various layers of IC device **2240** according to IC design layout diagram **2211**. Mask house **2220** performs mask data preparation **2221**, where IC design layout diagram **2211** is translated into a representative data file ("RDF"). Mask data preparation **2221** provides the RDF to mask fabrication **2222**. Mask fabrication **2222** includes a mask writer. A mask writer converts the RDF to an image on a substrate, such as a mask (reticle) **2223** or a semiconductor wafer **2233**. The IC design layout diagram **2211** is manipulated by mask data preparation **2221** to comply with particular characteristics of the mask writer and/or requirements of IC fab **2230**. In FIG. **22**, data preparation **2221** and mask fabrication **2222** are illustrated as separate elements. In some embodiments, data preparation **2221** and mask fabrication **2222** can be collectively referred to as mask data preparation.

(276) In some embodiments, data preparation **2221** includes optical proximity correction (OPC) which uses lithography enhancement techniques to compensate for image errors, such as those that can arise from diffraction, interference, other process effects and the like. OPC adjusts IC design layout diagram **2211**. In some embodiments, data preparation **2221** includes further resolution enhancement techniques (RET), such as off-axis illumination, sub-resolution assist features, phase-shifting masks, other suitable techniques, and the like or combinations thereof. In some embodiments, inverse lithography technology (ILT) is also used, which treats OPC as an inverse imaging problem.

(277) In some embodiments, data preparation **2221** includes a mask rule checker (MRC) that checks the IC design layout diagram **2211** that has undergone processes in OPC with a set of mask creation rules which contain certain geometric and/or connectivity restrictions to ensure sufficient margins, to account for variability in semiconductor manufacturing processes, and the like. In some

embodiments, the MRC modifies the IC design layout diagram **2211** to compensate for limitations during mask fabrication **2222**, which may undo part of the modifications performed by OPC in order to meet mask creation rules.

(278) In some embodiments, data preparation **2221** includes lithography process checking (LPC) that simulates processing that will be implemented by IC fab **2230** to fabricate IC device **2240**. LPC simulates this processing based on IC design layout diagram **2211** to create a simulated manufactured device, such as IC device **2240**. The processing parameters in LPC simulation can include parameters associated with various processes of the IC manufacturing cycle, parameters associated with tools used for manufacturing the IC, and/or other aspects of the manufacturing process. LPC takes into account various factors, such as aerial image contrast, depth of focus (“DOF”), mask error enhancement factor (“MEEF”), other suitable factors, and the like or combinations thereof. In some embodiments, after a simulated manufactured device has been created by LPC, if the simulated device is not close enough in shape to satisfy design rules, OPC and/or MRC are repeated to further refine IC design layout diagram **2211**.

(279) It should be understood that the above description of data preparation **2221** has been simplified for the purposes of clarity. In some embodiments, data preparation **2221** includes additional features such as a logic operation (LOP) to modify the IC design layout diagram **2211** according to manufacturing rules. Additionally, the processes applied to IC design layout diagram **2211** during data preparation **2221** may be executed in a variety of different orders.

(280) After data preparation **2221** and during mask fabrication **2222**, a mask **2223** or a group of masks **2223** are fabricated based on the modified IC design layout diagram **2211**. In some embodiments, mask fabrication **2222** includes performing one or more lithographic exposures based on IC design layout diagram **2211**. In some embodiments, an electron-beam (e-beam) or a mechanism of multiple e-beams is used to form a pattern on a mask (photomask or reticle) **2223** based on the modified IC design layout diagram **2211**. Mask **2223** can be formed in various technologies. In some embodiments, mask **2223** is formed using binary technology. In some embodiments, a mask pattern includes opaque regions and transparent regions. A radiation beam, such as an ultraviolet (UV) beam, used to expose the image sensitive material layer (for example, photoresist) which has been coated on a wafer, is blocked by the opaque region and transmits through the transparent regions. In one example, a binary mask version of mask **2223** includes a transparent substrate (for example, fused quartz) and an opaque material (for example, chromium) coated in the opaque regions of the binary mask. In another example, mask **2223** is formed using a phase shift technology. In a phase shift mask (PSM) version of mask **2223**, various features in the pattern formed on the phase shift mask are configured to have proper phase difference to enhance the resolution and imaging quality. In various examples, the phase shift mask can be attenuated PSM or alternating PSM. The mask(s) generated by mask fabrication **2222** is used in a variety of processes. For example, such a mask(s) is used in an ion implantation process to form various doped regions in semiconductor wafer **2233**, in an etching process to form various etching regions in semiconductor wafer **2233**, and/or in other suitable processes.

(281) IC fab **2230** includes wafer fabrication **2232**. IC fab **2230** is an IC fabrication business that includes one or more manufacturing facilities for the fabrication of a variety of different IC products. In some embodiments, IC Fab **2230** is a semiconductor foundry. For example, there may be a manufacturing facility for the front end fabrication of a plurality of IC products (front-end-of-line (FEOL) fabrication), while a second manufacturing facility may provide the back end fabrication for the interconnection and packaging of the IC products (back-end-of-line (BEOL) fabrication), and a third manufacturing facility may provide other services for the foundry business.

(282) IC fab **2230** uses mask(s) **2223** fabricated by mask house **2220** to fabricate IC device **2240**. Thus, IC fab **2230** at least indirectly uses IC design layout diagram **2211** to fabricate IC device **2240**. In some embodiments, semiconductor wafer **2233** is fabricated by IC fab **2230** using mask(s) **2223** to form IC device **2240**. In some embodiments, the IC fabrication includes performing one or

more lithographic exposures based at least indirectly on IC design layout diagram **2211**.

Semiconductor wafer **2233** includes a silicon substrate or other proper substrate having material layers formed thereon. Semiconductor wafer **2233** further includes one or more of various doped regions, dielectric features, multilevel interconnects, and the like (formed at subsequent manufacturing steps).

(283) Moreover, various circuits or devices to implement the transistors in the aforementioned embodiments are within the contemplated scope of the present disclosure. In some embodiments of this document, at least one of the transistors is implemented with at least one MOS transistor, at least one bipolar junction transistor (BJT), etc., or the combination thereof. Various circuits or devices to implement the transistors in the aforementioned embodiments are within the contemplated scope of the present disclosure.

(284) In some embodiments, an integrated circuit is disclosed and includes conductive rails that are disposed in a first conductive layer and separated from each other in a layout view, signal rails disposed in a second conductive layer different from the first conductive layer, at least one first via coupling a first signal rail of the signal rails to at least one of the conductive rails, and at least one first conductive segment. The first signal rail transmits a supply signal through the at least one first via and the at least one of the conductive rails to at least one element of the integrated circuit. The at least one first via and the at least one first conductive segment are disposed above first conductive layer. The at least one first conductive segment is coupled to the at least one of the conductive rails and is separate from the first signal rail.

(285) In some embodiments, the integrated circuit further includes at least one second conductive segment. The at least one second conductive segment is disposed between the first conductive layer and the second conductive layer. The at least one second conductive segment is disposed above and couples at least two adjacent conductive rails of the plurality of conductive rails to each other, in a layout view, and is separate from the plurality of signal rails.

(286) In some embodiments, the integrated circuit further includes at least one second via. The at least one second via is disposed between the first conductive layer and the second conductive layer. The at least one second conductive segment is coupled through the at least one second via to the plurality of signal rails. A height of the at least one second via is smaller than a height of the at least one first via.

(287) In some embodiments, the at least one first conductive segment includes separate portions, and the at least one first via is disposed between the separate portions, in a layout view. The at least one first via and the separate portions are disposed right above at least one of the plurality of conductive rails, in a layout view.

(288) In some embodiments, the integrated circuit further includes a plurality of second vias. The plurality of second vias are disposed between the first conductive layer and the second conductive layer. The at least one first conductive segment includes a pair of conductive segments. Each of the pair of conductive segments is disposed on two separate conductive rails of the plurality of conductive rails respectively and is coupled through one of the plurality of second vias to the plurality of signal rails.

(289) In some embodiments, a width of the at least one first conductive segment is smaller than or equal to a width of one of the plurality of conductive rails. A length of the at least one first conductive segment is smaller than or equal to a length of one of the plurality of conductive rails.

(290) In some embodiments, an integrated circuit is also disclosed. The integrated circuit includes a first plurality of conductive rails, a second plurality of conductive rails, at least one first via, and at least one first conductive segment. The first plurality of conductive rails are disposed in a first conductive layer and extend along a first direction. The first plurality of conductive rails are configured to transmit power signals to at least one element of the integrated circuit. The second plurality of conductive rails are disposed in the first conductive layer and extend along the first direction. The second plurality of conductive rails are disposed between the first plurality of

conductive rails and are separated from each other, in a layout view. The second plurality of conductive rails are configured to transmit data signals to the at least one element of the integrated circuit. The at least one first via is disposed between the first conductive layer and a second conductive layer above the first conductive layer, and couples one of the first plurality of conductive rails to a power rail disposed in the second conductive layer. The at least one first conductive segment is disposed between the first conductive layer and the second conductive layer above the first conductive layer. The at least one first conductive segment is at least partially overlapped with at least two adjacent conductive rails of the second plurality of conductive rails in a layout view, and the at least one first conductive segment contacts the at least two adjacent conductive rails of the second plurality of conductive rails and is separated from a signal rail that is disposed in the second conductive layer and next to the power rail. The at least one first conductive segment extends along the first direction.

(291) In some embodiments, the integrated circuit further includes a plurality of second conductive segments. The plurality of second conductive segments are disposed between the first conductive layer and the second conductive layer. The plurality of second conductive segments extend along the first direction and are disposed right above the first plurality of conductive rails in a layout view. The plurality of second conductive segments does not exceed the first plurality of conductive rails in a layout view, and are separated from the second conductive layer.

(292) In some embodiments, one of the plurality of second conductive segments includes separate portions, and the at least one first via is disposed between the separate portions, in a layout view.

(293) In some embodiments, the integrated circuit further includes at least one second via. The at least one second via is disposed between the first conductive layer and the second conductive layer. One of the plurality of second conductive segments is coupled through the at least one second via to the power rail. A height of the at least one first via is different from a height of the at least one second via.

(294) In some embodiments, the integrated circuit further includes a third conductive segment and a fourth conductive segment. The third conductive segment is disposed over an active area below the first conductive layer and extends along a second direction, for forming a first transistor. The third conductive segment is overlapped with a first conductive rail of the second plurality of conductive rails in a layout view. The fourth conductive segment is disposed over the active area and extends along the second direction, for forming a second transistor that couples to the first transistor. The fourth conductive segment is overlapped with a second conductive rail of the second plurality of conductive rails in a layout view. The first conductive rail and the second conductive rail are next to each other, and the first conductive rail, the second conductive rail, and the at least one first conductive segment are partially overlapped together in a layout view.

(295) In some embodiments, the integrated circuit further includes at least one via. The at least one via is disposed below the first conductive layer. The at least one second via is at least overlapped with the second plurality of conductive rails in a layout view. The at least one first conductive segment is overlapped with the at least one second via, the at least two adjacent conductive rails of the second plurality of conductive rails, and a cut portion of the at least two adjacent conductive rails, in a layout view.

(296) In some embodiments, the integrated circuit further includes at least one third conductive segment. The at least one third conductive segment is disposed over an active area below the first conductive layer and extends along a second direction, for forming gate terminals. The at least one third conductive segment is disposed across the second plurality of conductive rails, in a layout view. The at least one third conductive segment is coupled through the at least one second via to one of the second plurality of conductive rails.

(297) In some embodiments, the at least one third conductive segment includes a plurality of third conductive segments. The plurality of third conductive segments are separated from each other in the first direction. A length of the at least one first conductive segment is substantially equal to or

greater than a distance between two adjacent third conductive segments of the plurality of third conductive segments.

(298) In some embodiments, the integrated circuit further includes a plurality of third conductive segments and at least one second via. The plurality of third conductive segments are disposed over an active area below the first conductive layer and extend along a second direction, for forming transistors. The at least one second via is disposed below the first conductive layer. One of the plurality of third conductive segments is coupled through the at least one second via to one of the second plurality of conductive rails that is coupled through the at least one first via to the power rail. A height of the at least one first via is greater than a height of at least one first conductive segment.

(299) In some embodiments, an integrated circuit is provided and includes a plurality of conductive rails in a first layer, a first conductive structure surrounding by a patterned film structure in a second layer above the first layer, wherein the first conductive structure contacts a first conductive rail of the plurality of conductive rails; a dielectric structure in a third layer to cover the first conductive structure; and a second conductive structure that includes a signal rail and a first via that passes through the patterned film structure and the dielectric structure and is coupled between the signal rail and a second conductive rail of the plurality of conductive rails.

(300) In some embodiments, the integrated circuit further includes a second via that passes through the dielectric structure to contact the first conductive structure and further contacts a power rail disposed next to the second conductive rail.

(301) In some embodiments, the integrated circuit further includes a third conductive structure contacting two adjacent conductive rails of the plurality of conductive rails. The third conductive structure is separated from the plurality of conductive rails.

(302) In some embodiments, the integrated circuit further includes a second via that contacts the first conductive rail and a power rail disposed next to the signal rail. the first conductive structure comprises separate segments, and the second via is disposed between the separate segments of the first conductive structure.

(303) In some embodiments, a thickness of the film structure is substantially equal to a distance between the second layer and the third layer.

(304) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. An integrated circuit, comprising: a plurality of conductive rails that are disposed in a first conductive layer and separated from each other in a layout view; a first conductive segment coupled and arranged on first and second rails in the plurality of conductive rails; a trapezium structure arranged on the first conductive segment and overlapping the first and second rails in the layout view; and a first signal rail disposed in a second conductive layer above the first conductive layer and arranged on the trapezium structure, wherein the first signal rail is configured to transmit a data signal to the first and second rails.
2. The integrated circuit of claim 1, the trapezium structure has a first surface contacting the first signal rail a second surface contacting the first conductive segment, wherein in a cross-section view a width of the first surface is greater than a width of the second surface.

3. The integrated circuit of claim 1, further comprising: a second conductive segment arranged on a third rail in the plurality of conductive rails, wherein a height of the second conductive segment is smaller than a sum of a height of the first conductive segment and a height of the trapezium structure.
4. The integrated circuit of claim 3, wherein in a cross-section view a width of the third rail is greater than a width of the first and second rails.
5. The integrated circuit of claim 1, further comprising: a second conductive segment arranged below the first and second rails; and a via coupled between the second conductive segment and the second rail.
6. The integrated circuit of claim 1, wherein the first signal rail is arranged perpendicular to the first and second rails.
7. An integrated circuit, comprising: a plurality of first conductive rails and a plurality of second conductive rails that are disposed in a first conductive layer and extend in a first direction; first and second conductive segments disposed on and coupled to the plurality of first conductive rails; a first conductive structure that is interposed between the first and second conductive segments along a second direction and arranged on the plurality of second conductive rails; and a first signal rail disposed in a second conductive layer above the first conductive layer, and coupled to the plurality of second conductive rails through the first conductive structure, wherein a height of the first and second conductive segments is smaller than a height of the first conductive structure.
8. The integrated circuit of claim 7, further comprising: a via disposed on the first conductive segment to form a second conductive structure that is coupled to a first rail of the plurality of first conductive rails, wherein a height of the second conductive structure is the same as the height of the first conductive structure.
9. The integrated circuit of claim 8, wherein a width of the plurality of first conductive rails is greater than a width of the plurality of second conductive rails along the second direction.
10. The integrated circuit of claim 8, further comprising: a second signal rail disposed in the second conductive layer and coupled to the first rail through the second conductive structure.
11. The integrated circuit of claim 10, wherein the first and second signal rails extend in the second direction to pass the plurality of first conductive rails and the plurality of second conductive rails.
12. The integrated circuit of claim 7, further comprising: a third conductive segment arranged below the plurality of first conductive rails and the plurality of second conductive rails; and a via coupled between the third conductive segment and one of the plurality of second conductive rails.
13. The integrated circuit of claim 7, wherein in a layout view a width of the first and second conductive segment is greater than a width of the first conductive structure along the first direction.
14. The integrated circuit of claim 13, wherein in the layout view the width of the first conductive structure is greater than a width of the first signal rail along the first direction.
15. The integrated circuit of claim 7, wherein in a layout view the plurality of second conductive rails are arranged between the plurality of second conductive rails.
16. An integrated circuit, comprising: a plurality of first conductive rails in a first conductive layer; a first conductive segment on first and second rails in the plurality of conductive rails; a first trapezium structure on the first conductive segment and above the first and second rails, wherein along a first direction a width of the first trapezium structure is larger than a width of the first conductive segment, and along a second direction a length of the first trapezium structure is larger than a length of the first conductive segment; and a first signal rail in a second conductive layer above the first conductive layer and on the first trapezium structure.
17. The integrated circuit of claim 16, wherein the first signal rail extends to pass the plurality of first conductive rails.
18. The integrated circuit of claim 16, further comprising: a plurality of second conductive rails on opposite sides of the plurality of first conductive rails, wherein a width of the plurality of second conductive rails is greater than a width of the plurality of first conductive rails.

19. The integrated circuit of claim 18, further comprising: a third conductive segment on one of the plurality of second conductive rails; a second trapezium structure on the third conductive segment; and a second signal rail on the second trapezium structure.

20. The integrated circuit of claim 16, wherein a distance between the first signal rail and the plurality of first conductive rails is greater than a height of the first trapezium structure.
